

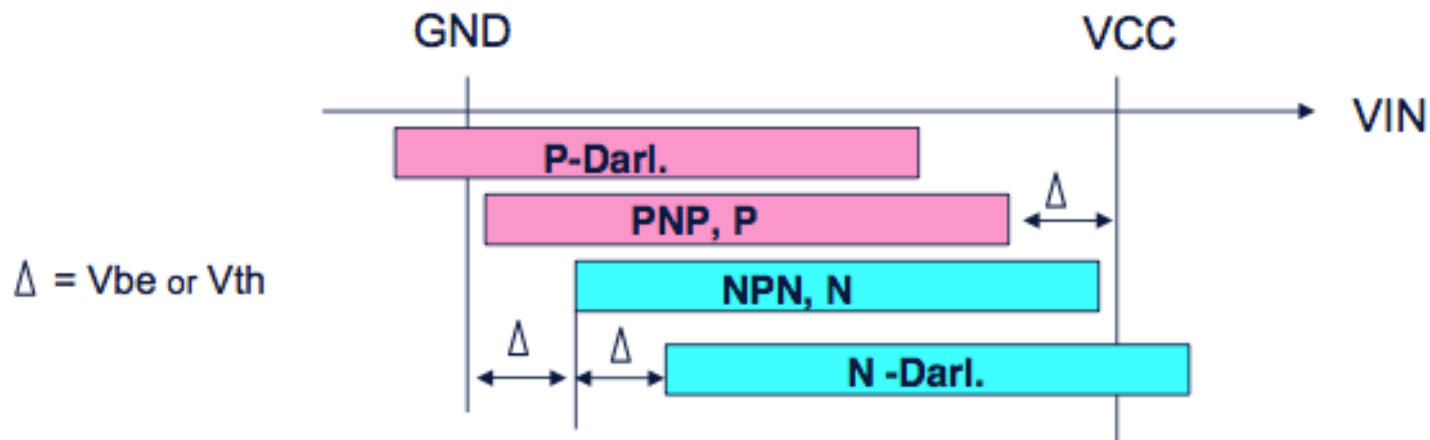
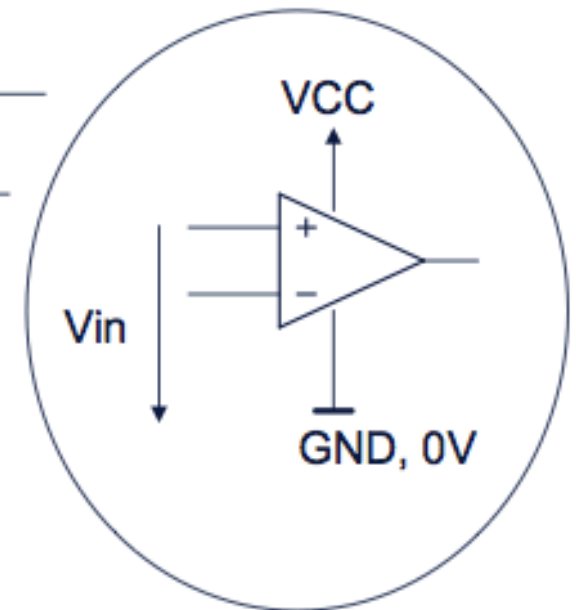
Bipolar (BJT) or MOSFET ?

	bip	MOS
Accuracy	+	-
Offset	+	-
Noise	+	-
Process deviation	+	-
Chip area	-	+
Voltage capability	-	+
Robust at high currents (power)	-	+
Current consumption	-	+
Revers-current sensitivity	-	+

N- or P-Input for Op-Amp ?

Choose topologie according to input voltage range

min	0 V	pnp, p-channel input
min	< 0 (-0.x) V	darlington pnp, p-channel
min	> 0,7V bzw. > V_{th}	nnp, n-channel
max	VCC	nnp, n-channel
max.	> VCC + 0.x V	darlington npn, ..



Comparator, Opamp, OTA ?

Comparator

The general circuit which is based on a differential-pair input to compare voltages.
Output stage can be digital.

Opamp, Operational Amplifier

If you use a comparator in a feedback loop, you need a well known frequency response/open loop gain characteristic.

You should add internal capacitors which are not required for a comparator.

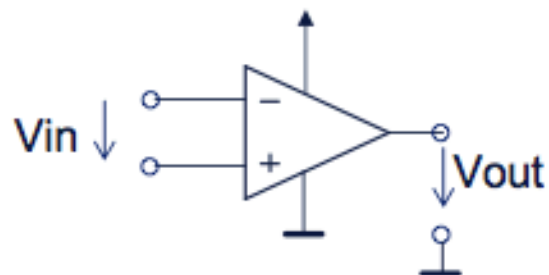
OTA , Operational Transconductance Amplifier

If you take it very seriously, you should say „OTA“ to a simple MOS amplifier because it does not have a high voltage gain,
but a high transconductance = ratio of output current / input voltage

MOS-opamp = OTA + output buffer

Comparator, Opamp, OTA ?

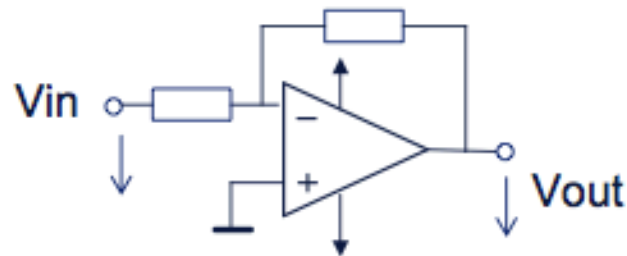
comparator



no feedback:
high voltage gain, no
special frequency response
required for stability if no
feedback loop

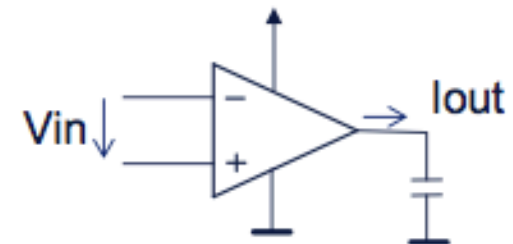
$$V_{out} = A \cdot V_{in}$$

op-amp operational amplifier



for feedback loop:
check frequency response
for stability: internal or
external frequency comp.
components (C,R) required.
output must be able to drive
ohmic loads

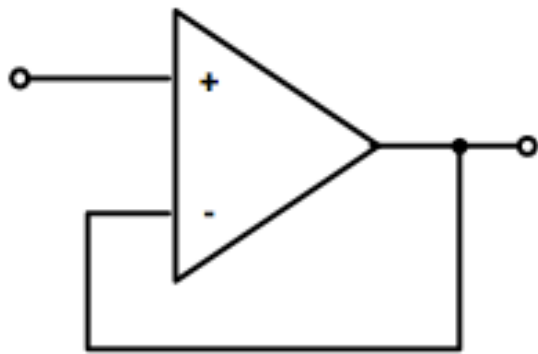
OTA operational trans- conductance amplifier



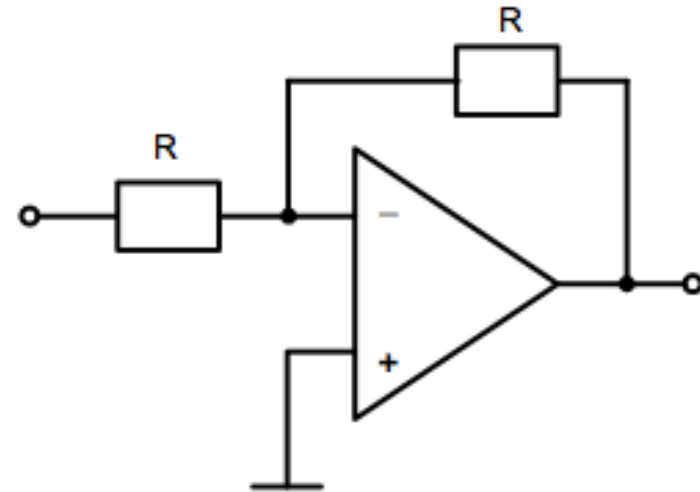
input voltage is translated
to output current:

$$I_{out} = G_m \cdot V_{in}$$

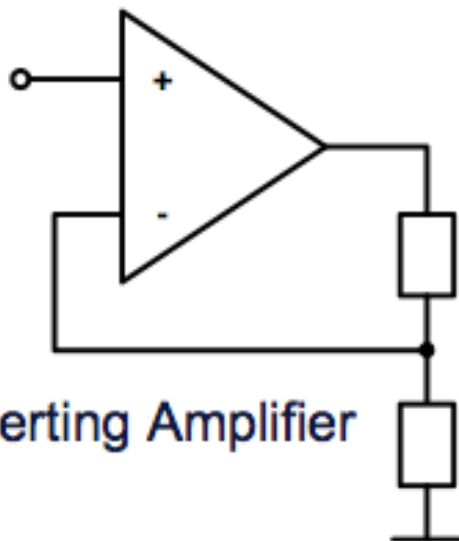
Basic use of OpAmps



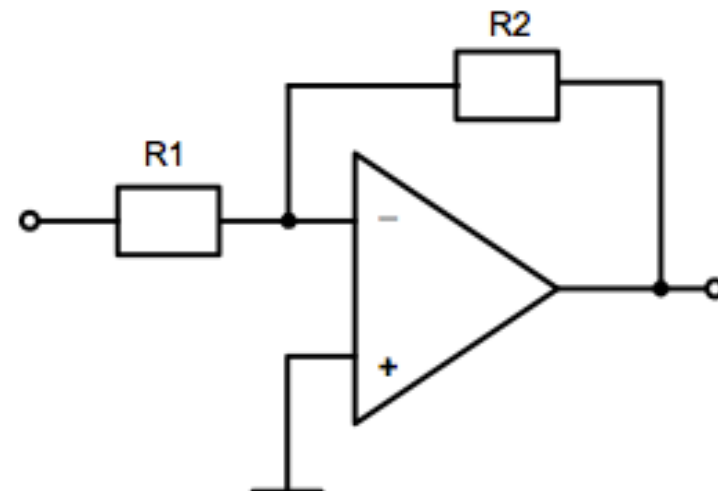
Noninverting Buffer



Inverting Buffer

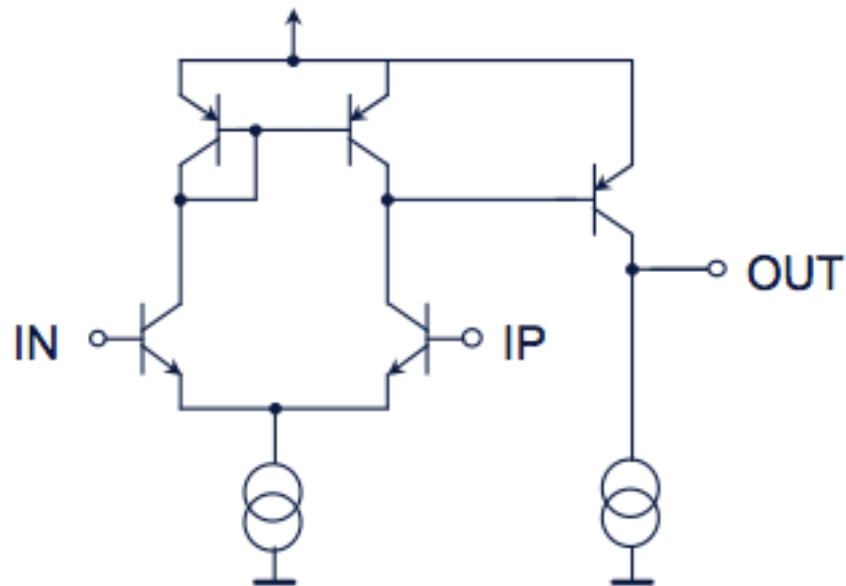


Noninverting Amplifier

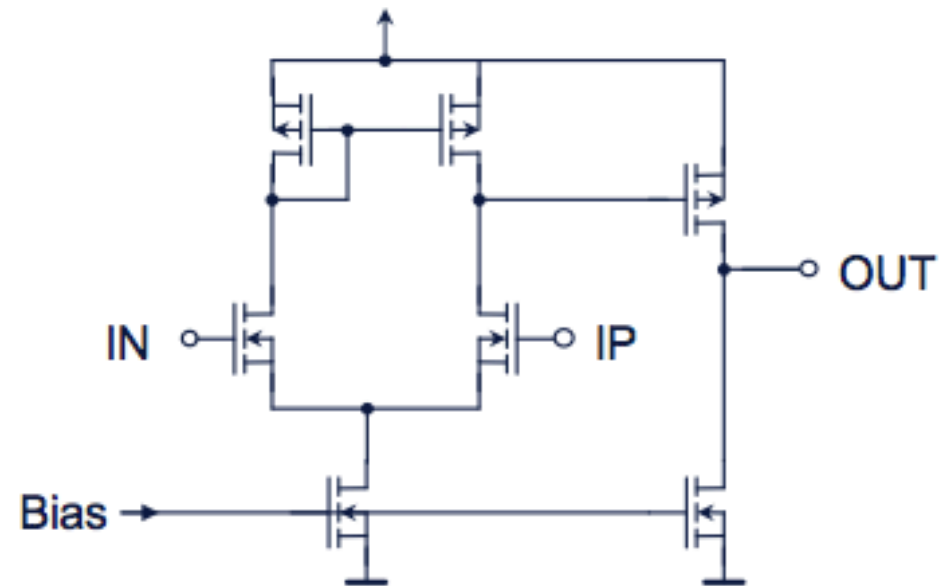


Inverting Amplifier

Comparator with N-device as input



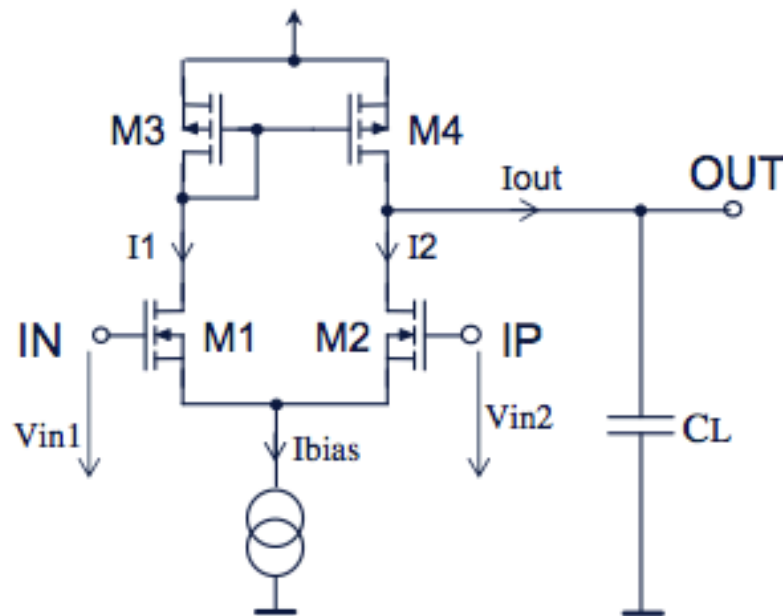
bipolar: npn-input



MOS: n-channel input

bipolar or MOS: very similar circuit topologies can be used for simple comparators

Simple 1-stage CMOS - OTA



$$\text{Transconductance } g_m = \frac{I_{OUT}}{(V_{IN1} - V_{IN2})}$$

Current mirror M3/M4 transfers differential input to single output

Recommended rules for transistor sizing:

M1,M2: choose large w for high gain; large area ($w \cdot l$) for low random offset

M3,M4: choose large l for high gain and low systematic offset

g_m – compare BJT / MOSFET

Bipolar: g_m increases linear with current

$$I_C = I_S \cdot e^{\frac{V_{BE}}{V_t}}$$

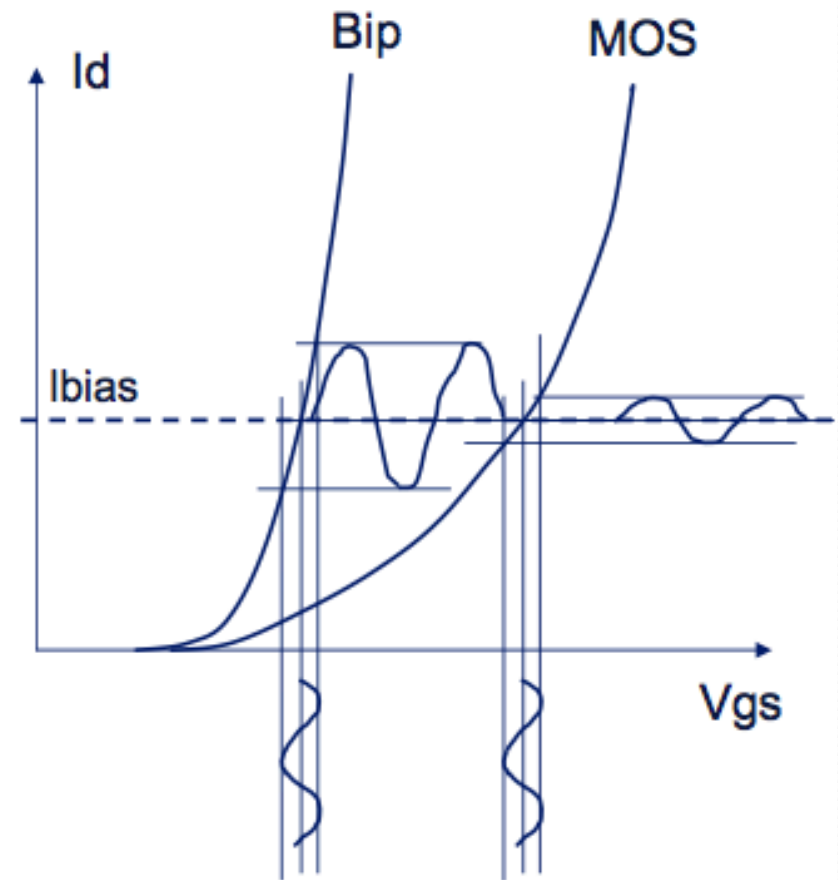
$$g_m = \frac{d I_C}{d V_{BE}} = \frac{I_C}{V_t}$$

MOS: g_m increases with squareroot of current

$$I_D = k \cdot \frac{w}{l} \cdot (V_{GS} - V_{th})^2$$

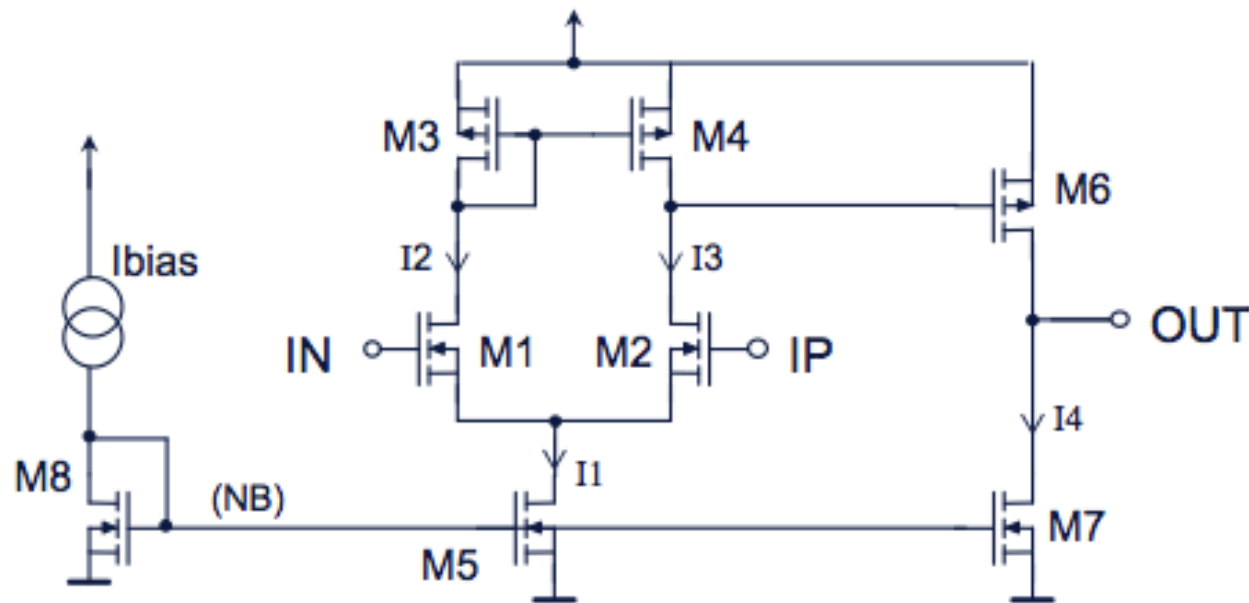
$$g_m = \frac{d I_D}{d V_{GS}} = k \cdot \frac{w}{l} \cdot 2 \cdot (V_{GS} - V_{th})$$

$$g_m = 2 \cdot \sqrt{k \cdot \frac{w}{l} \cdot I_D}$$



bipolar transistor will achieve more g_m

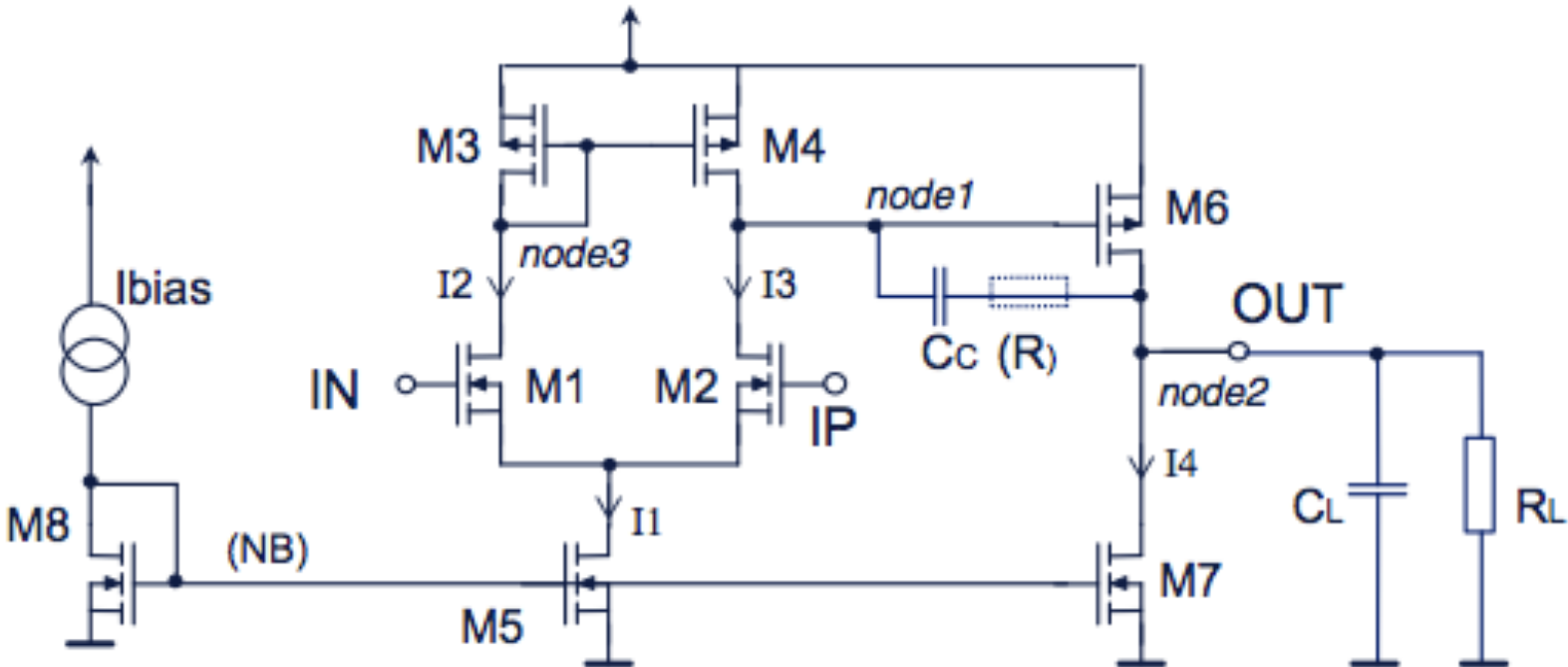
2-stage MOSFET – Comparator



Adding a second stage increases the gain and allows full swing at the output

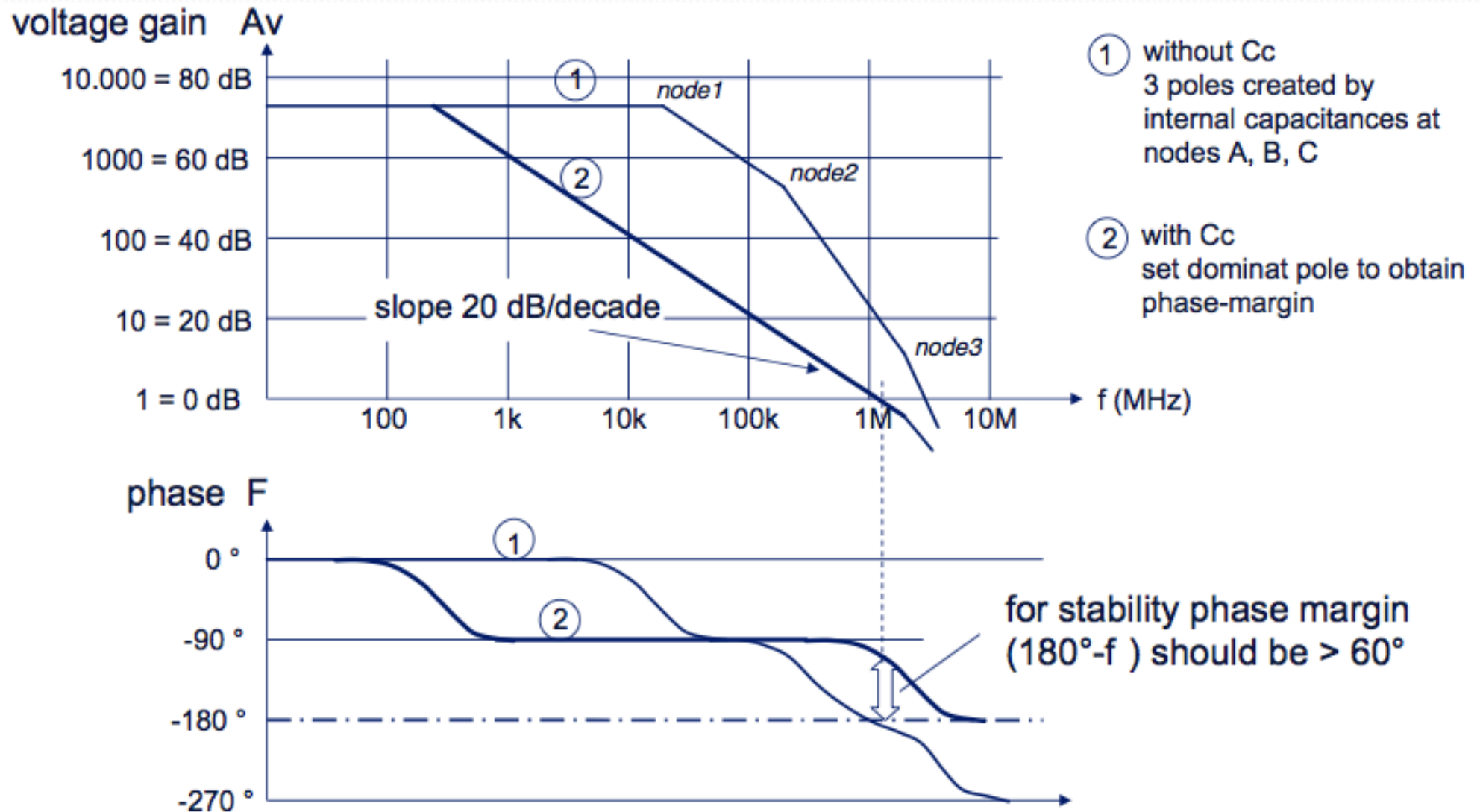
For use as a comparator (without feedback loop): no compensation C required

2-stage MOSFET – OTA „Miller OTA“

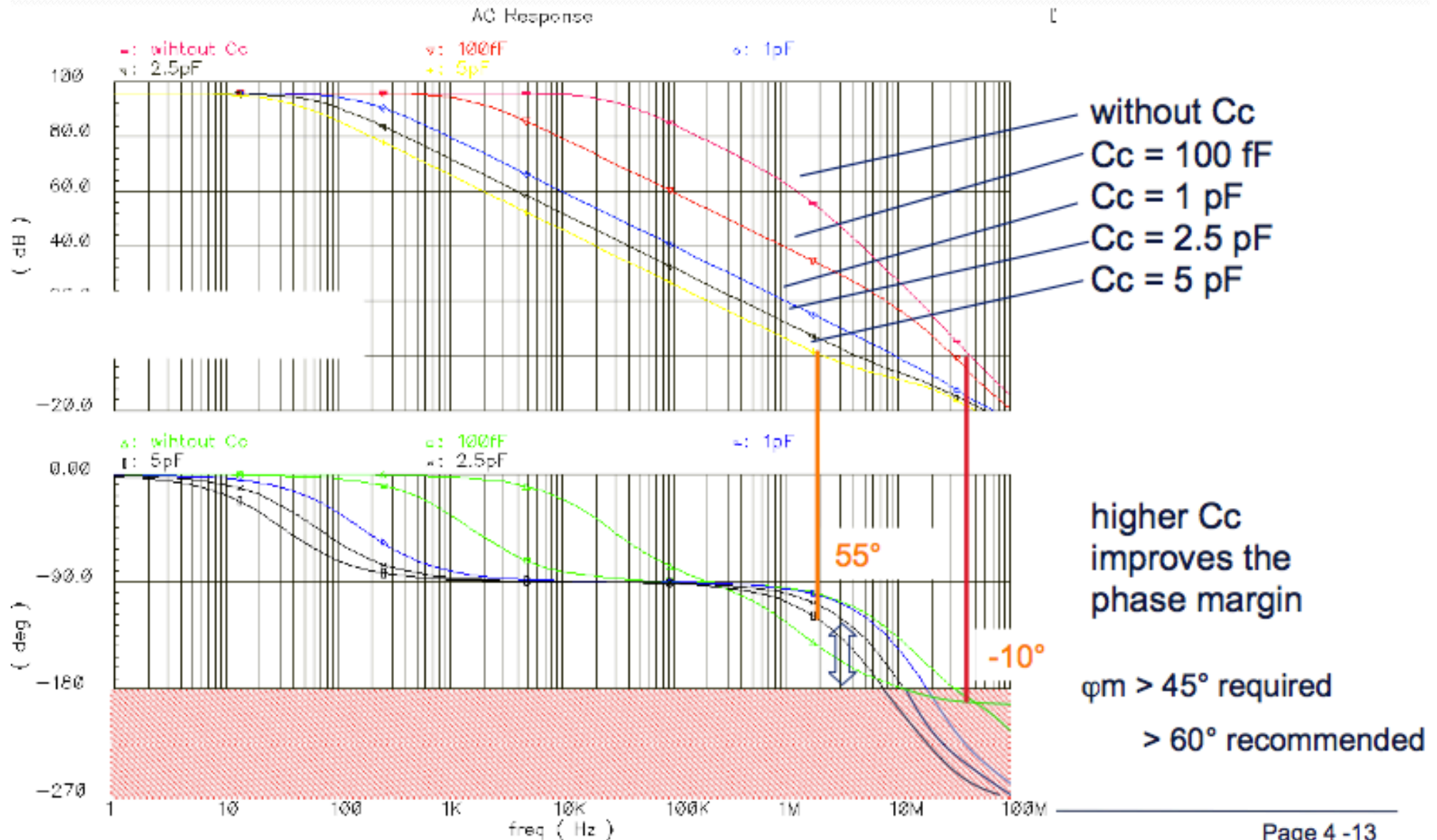


For using the 2-stage OTA in a feedback loop add a compensation-capacitor C_c or a compensation network $R + C$

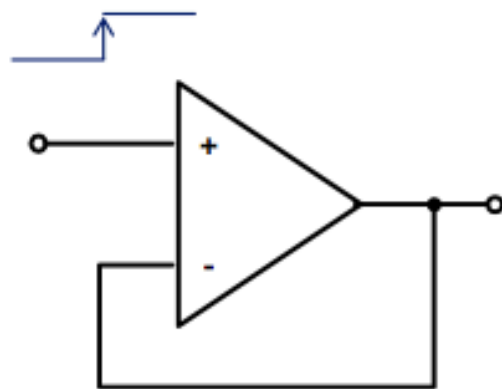
Bode diagram of Miller OTA



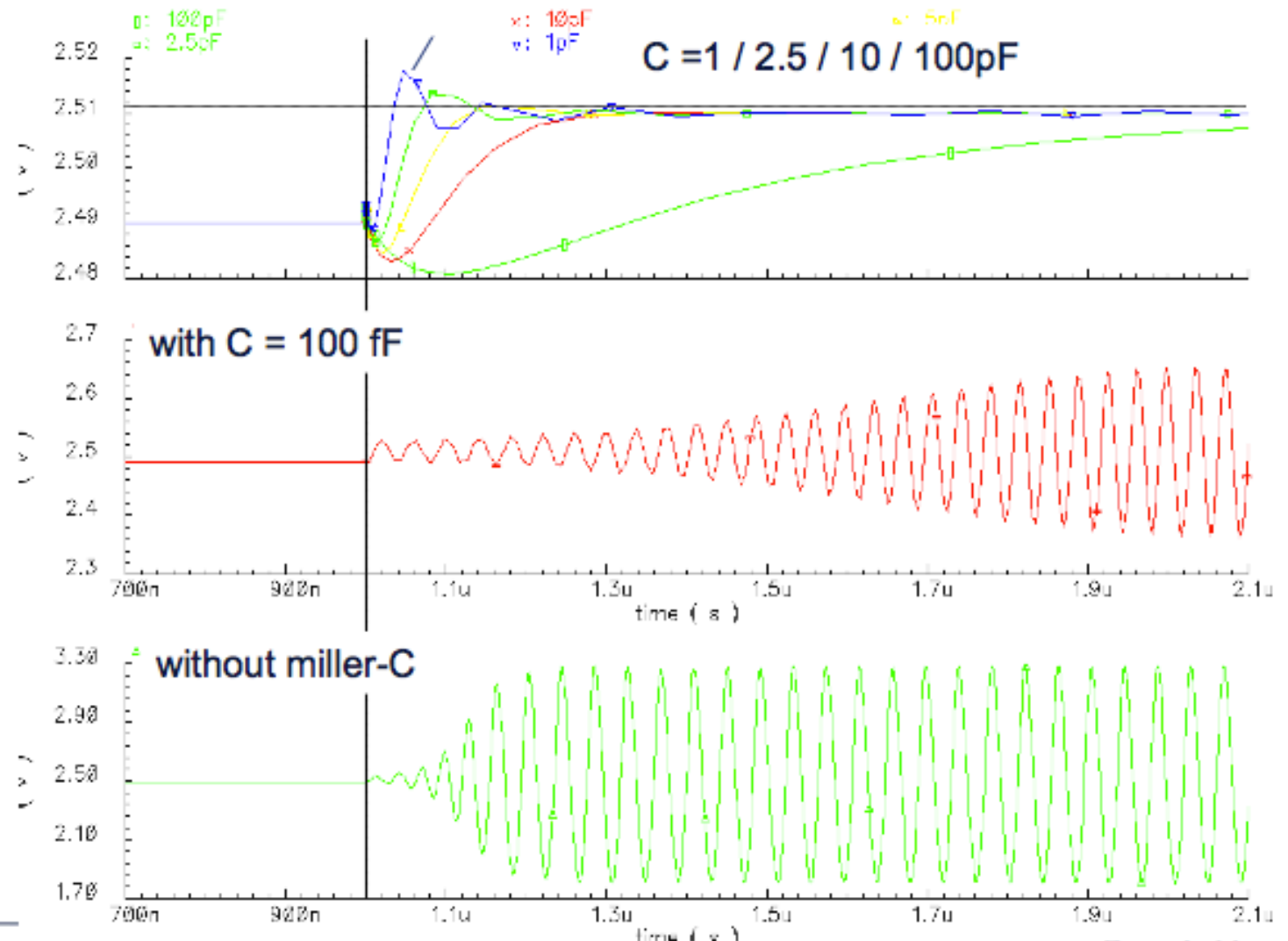
AC-Simulation of a 2-stage Miller OTA



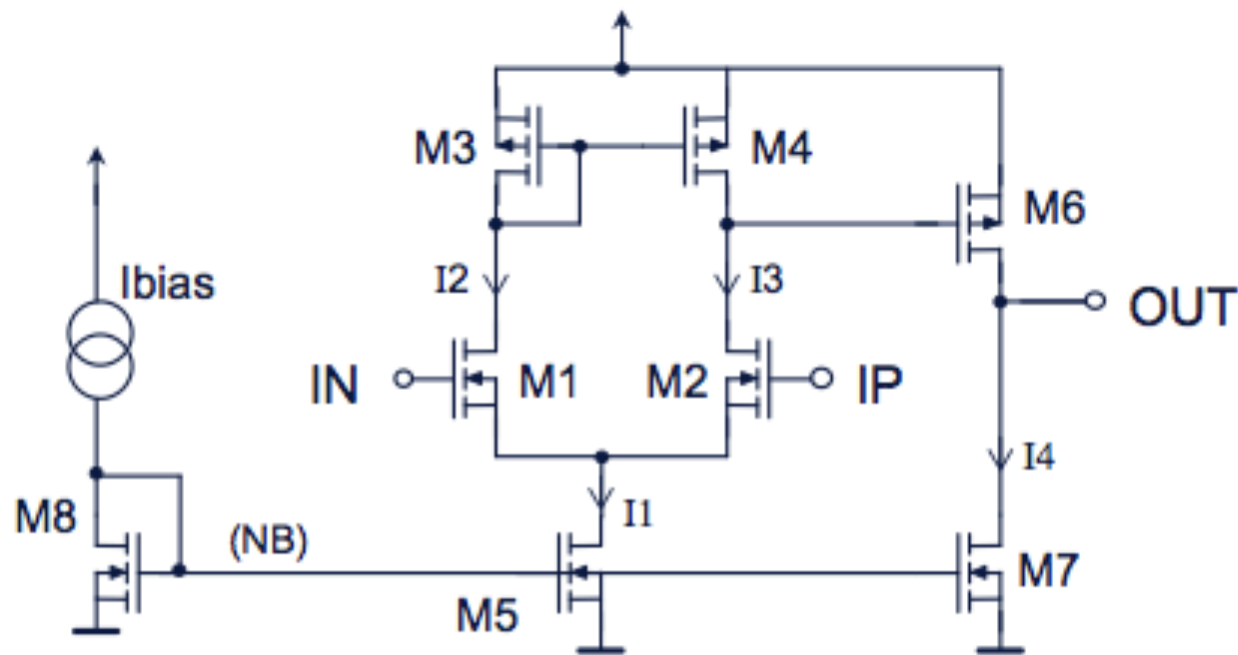
AC-analysis: Check Stability with Transient Analysis



Noninverting Buffer



w/o scaling for 2-stage MOSFET - comparator



$$(w/l)_1 = (w/l)_2$$

$$(w/l)_3 = (w/l)_4$$

at switching point
($V_{IP}=V_{IN}$):

$$I_2 = I_3 = \frac{I_1}{2}$$

to avoid systematic
offset choose:

$$\frac{(w/l)_6}{(w/l)_4} = 2 \cdot \frac{(w/l)_7}{(w/l)_5}$$

For MOS comparators/opamps this rule helps to avoid systematic offset, caused by the limited gain of the first stage

How to avoid/to minimize offset of a comparator

Random offset:

Use large devices

Use same size devices for matching, do not match small with large devices

Same orientation

Far away from heating sources (if possible) or on isotherms (line of same temp.)

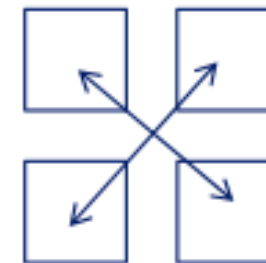
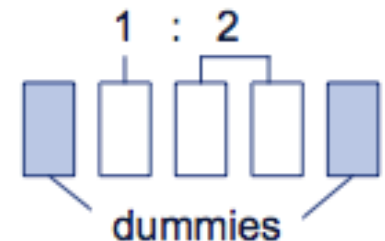
Symmetric wiring (poly, alu)

Avoid other wiring with high voltages over opamps layout

Use *dummy devices* to improve matching

Try a **centroid layout** for the devices to be matched:

both transistors of a differential pair are
doubled and connected pairwise in parallel
this helps to average out global errors

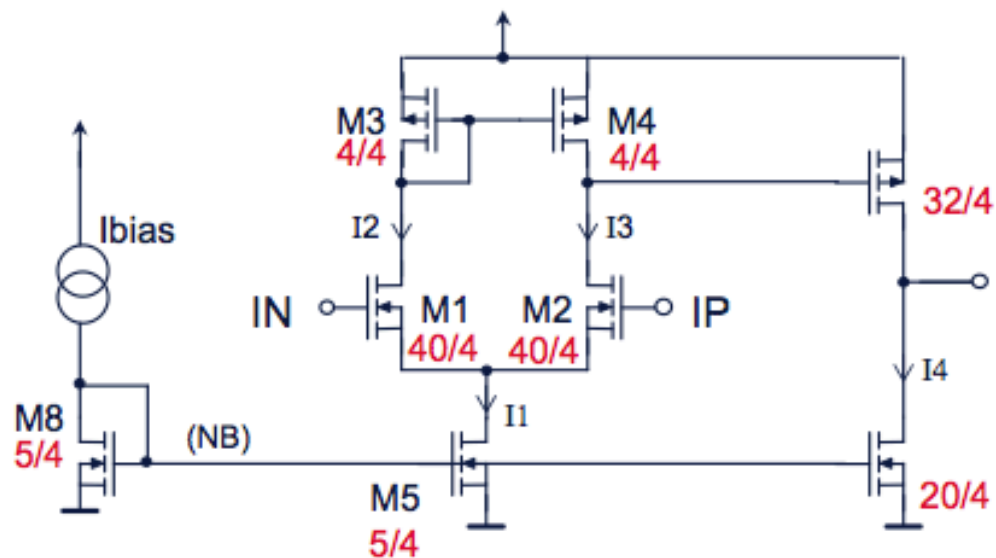


Systematic offset:

Use symmetry insofar as possible

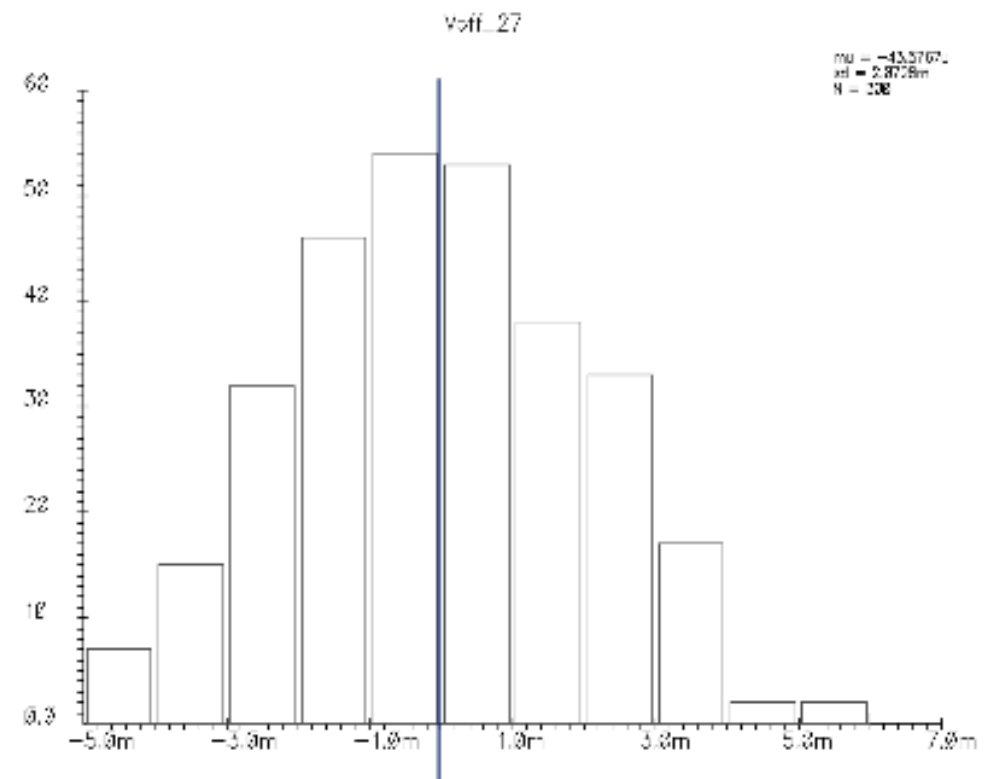
Most offset is caused by first stages, but do not forget the second stage

Random Offset – Monte Carlo Simulation



for MOS comparators you have to accept a
offset in the xx mV range,
use MC-simulation to get a better feeling

(simulation done with models of 0.5 μ m technology
with gateoxid = 15nm)

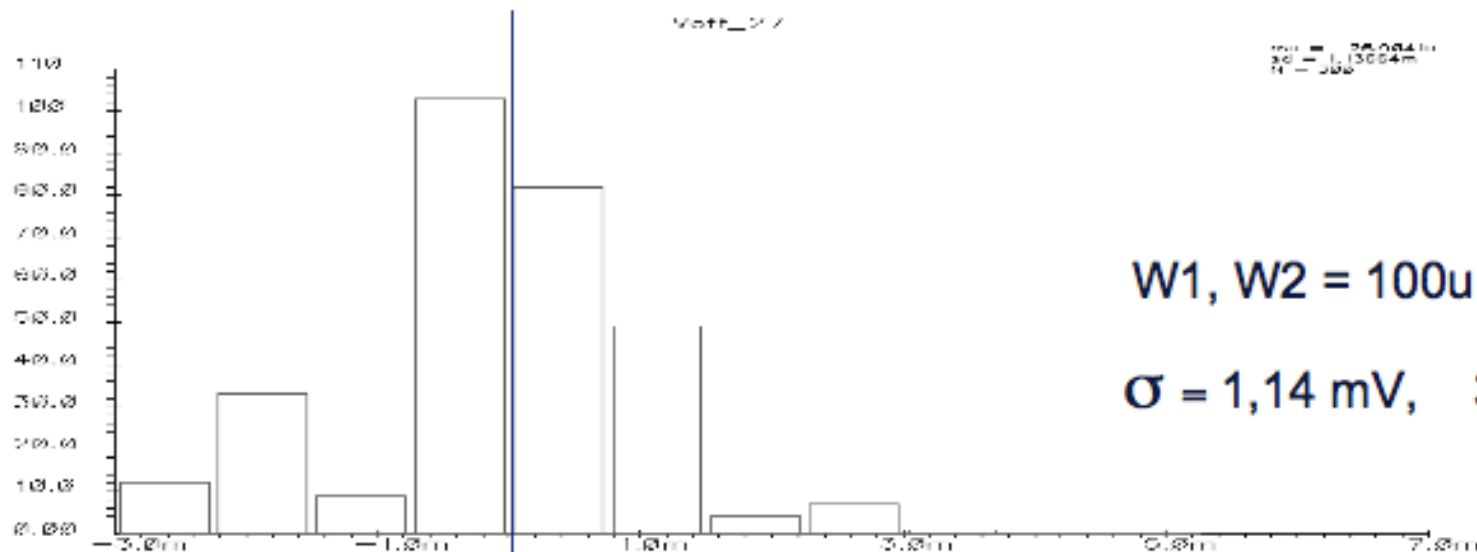


offset simulation:

standard deviation $\sigma = 2\text{mV}$, $3\sigma = 6\text{mV}$

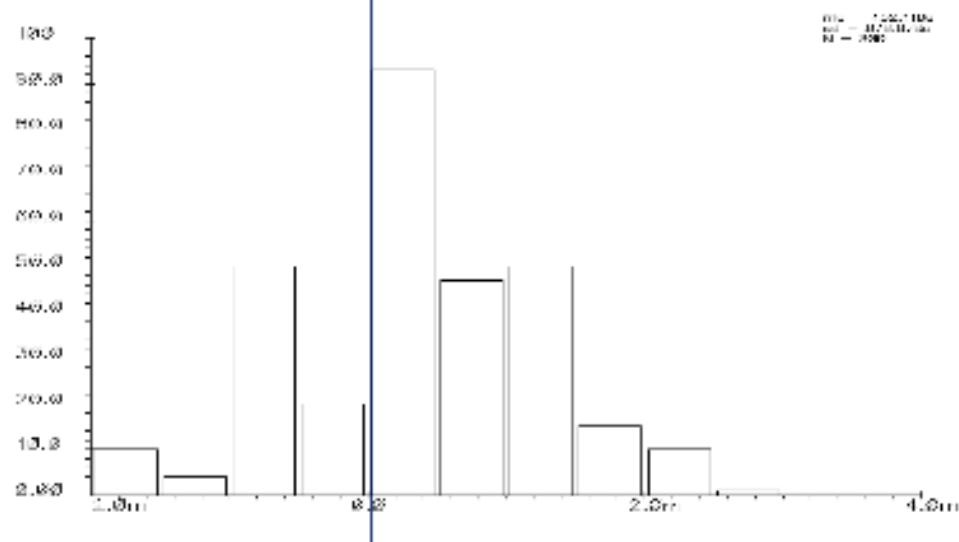
good centered, no systematic offset

Improve the offset with larger devices



W1, W2 = 100u

$\sigma = 1,14 \text{ mV}$, $3\sigma = 3,42 \text{ mV}$

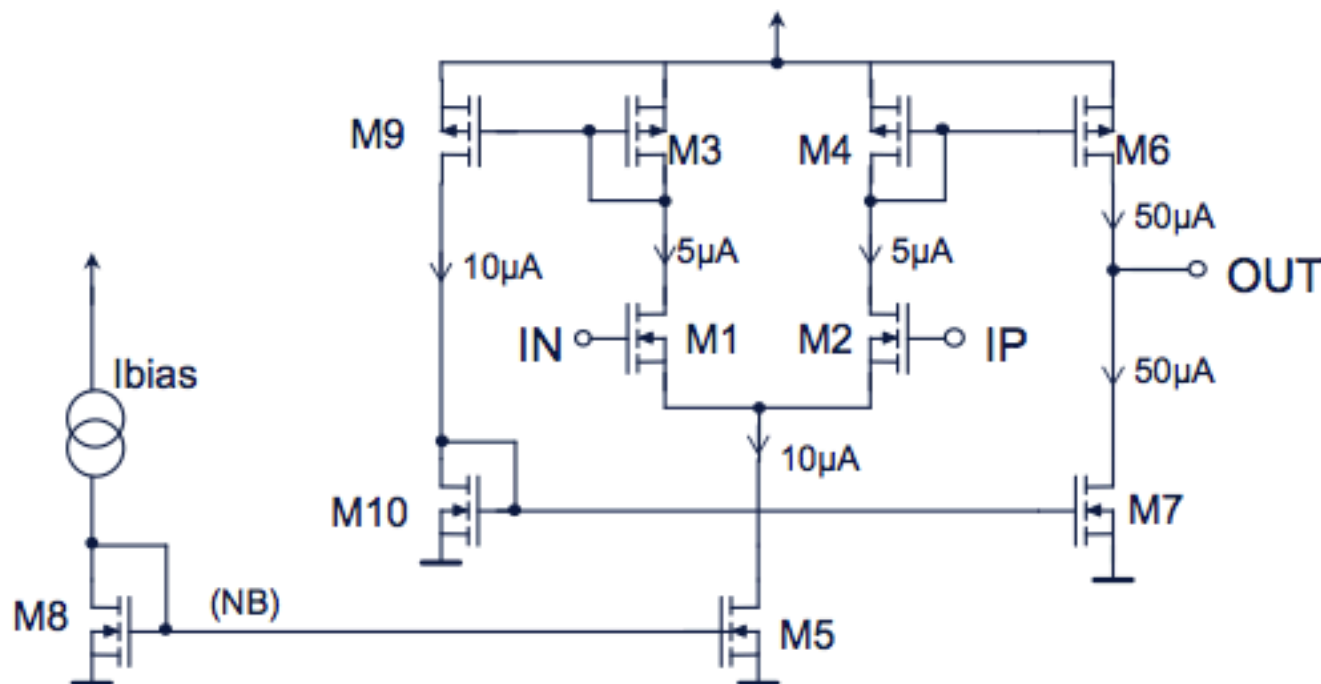


W1, W2 = 200u

$\sigma = 0,88 \text{ mV}$, $3\sigma = 2,64 \text{ mV}$

Offset improves approx. with the square-root of the area

MOSFET - comparator with push-pull current output



Rule for no systematic offset:

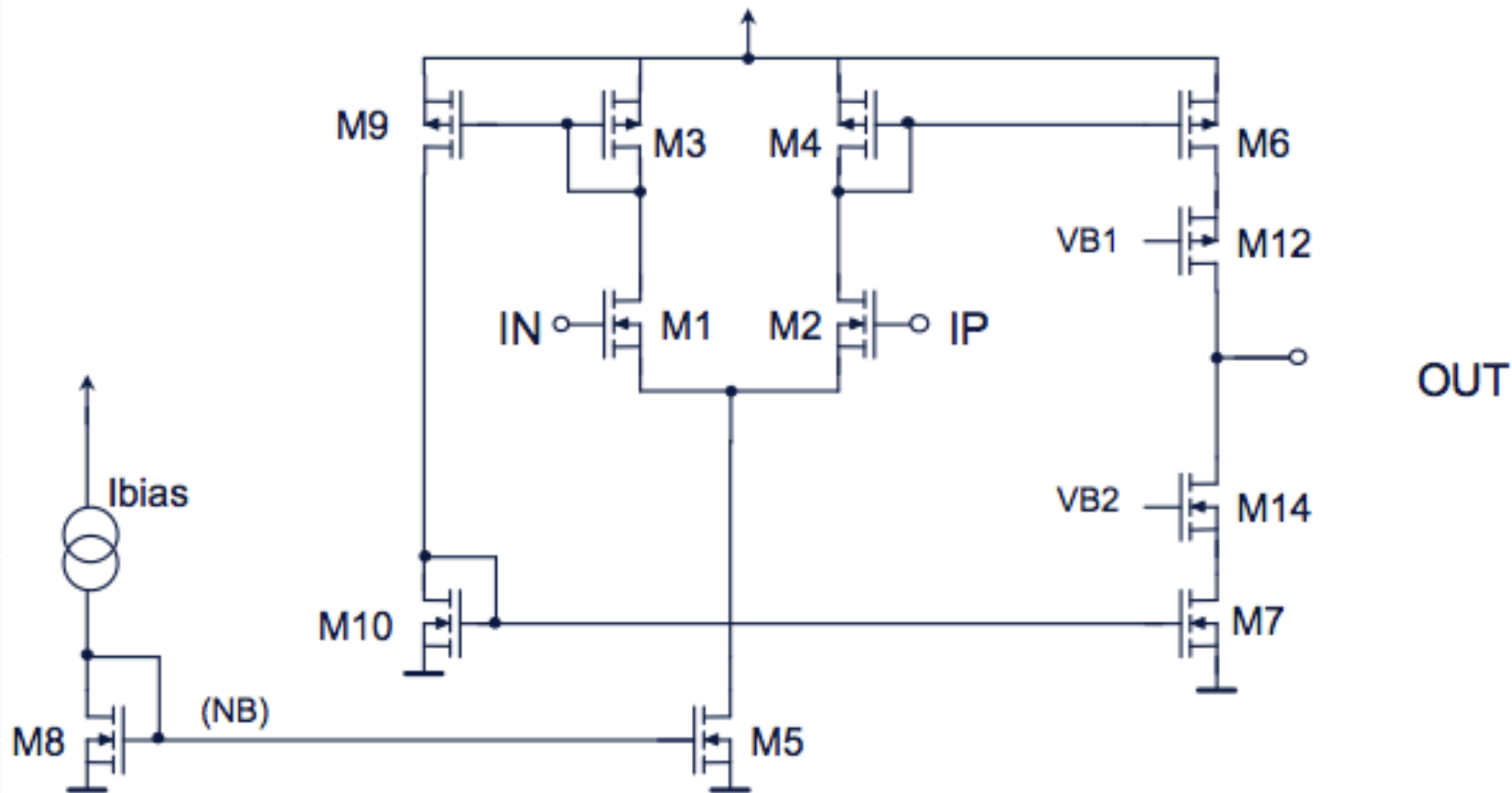
$$\frac{(w/l)_6}{(w/l)_4} = \frac{(w/l)_7}{(w/l)_{10}} \cdot \frac{(w/l)_9}{(w/l)_3}$$

The symmetrical design is a method to achieve low systematic offset

According to voltage gain it is rated as a „single-stage“ amplifier, because the output stage is only a current mirror, no gain stage.

This low gain also simplifies stability consideration in feedback loops.

MOSFET - comparator (OTA) with cascode output

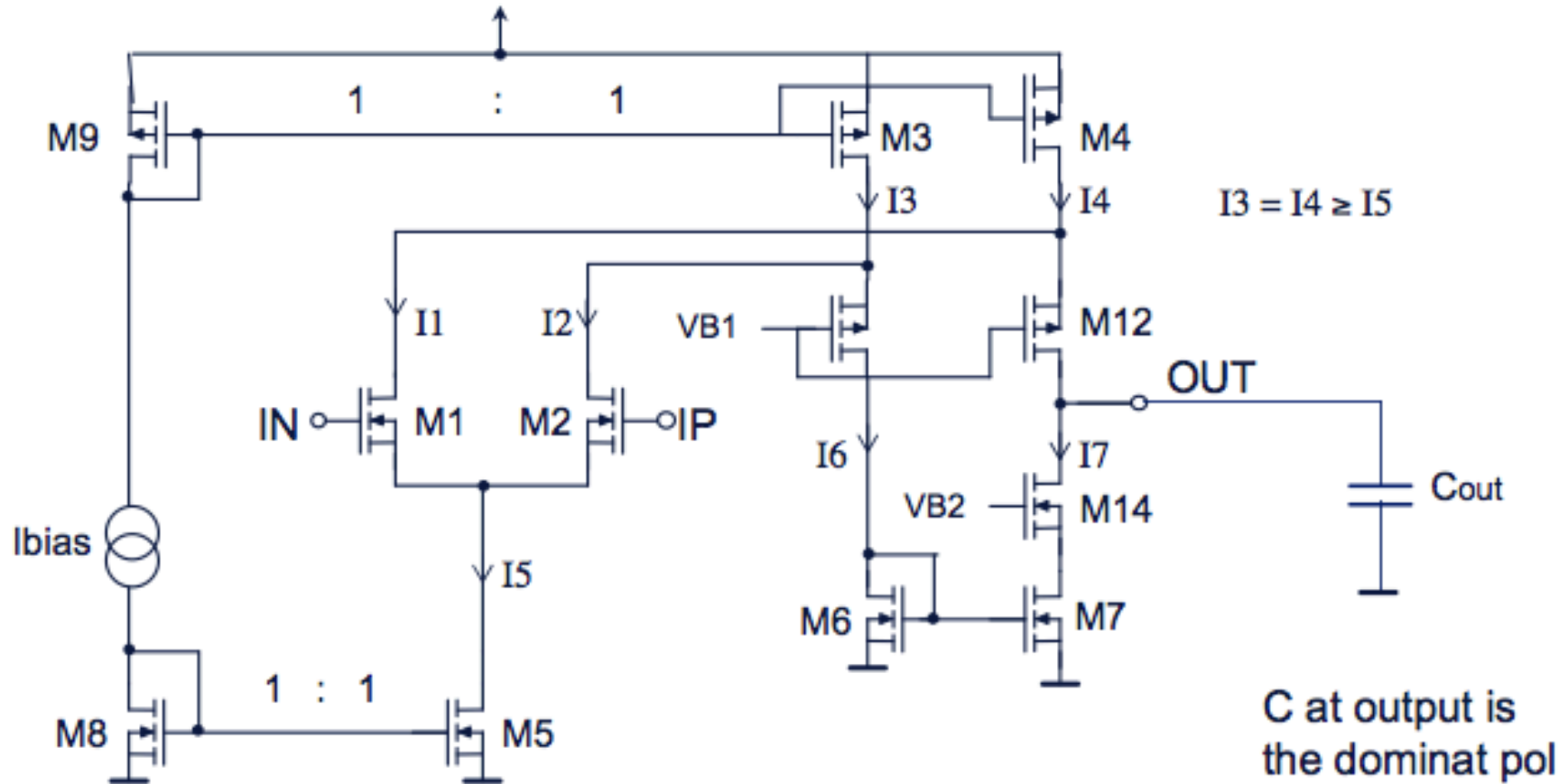


Cascode output is a method to increase open-loop gain without adding a second stage

$VB1, VB2$:
Bias voltages for cascode

Disadvantage: Limited voltage swing at output

MOSFET - OTA: Folded cascode

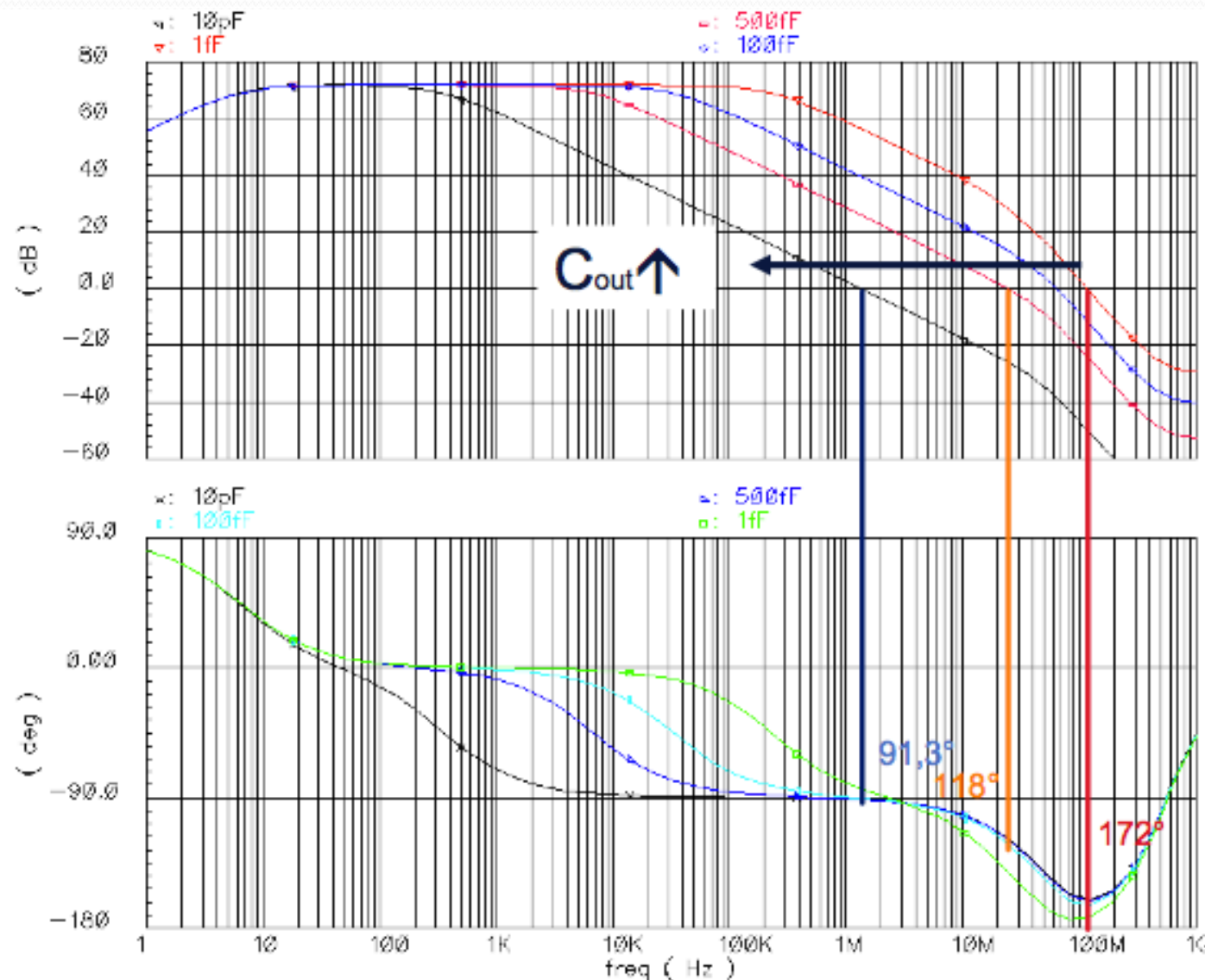


static condition: $I_3 = I_5 = 2 * I_2 \rightarrow I_6 = I_2, I_7 = I_1$

Advantage against normal cascode output (page before): extended output swing

Single gain stage, therefore no internal compensation capacitance is required

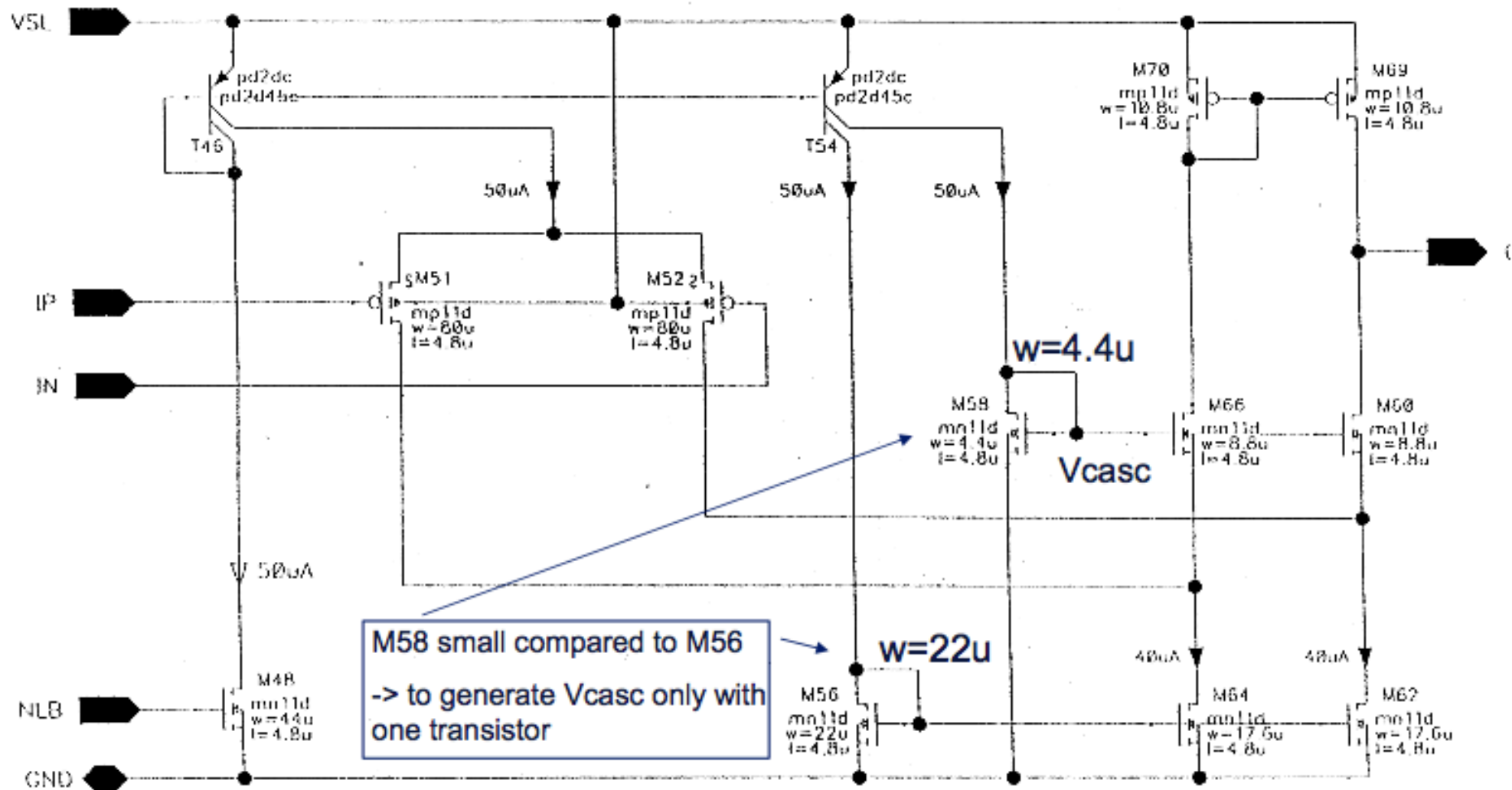
Folded cascode OTA – frequency response



larger C_{out}
increases the
phase margin

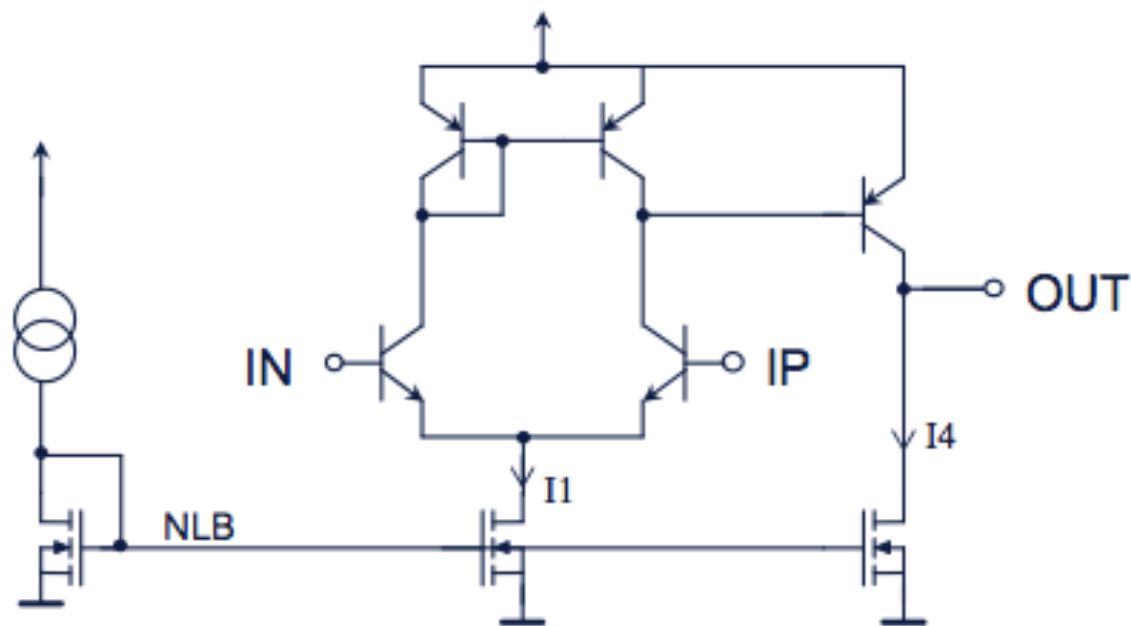
no internal
compensation-C
required

Folded Cascode example with p-channel input



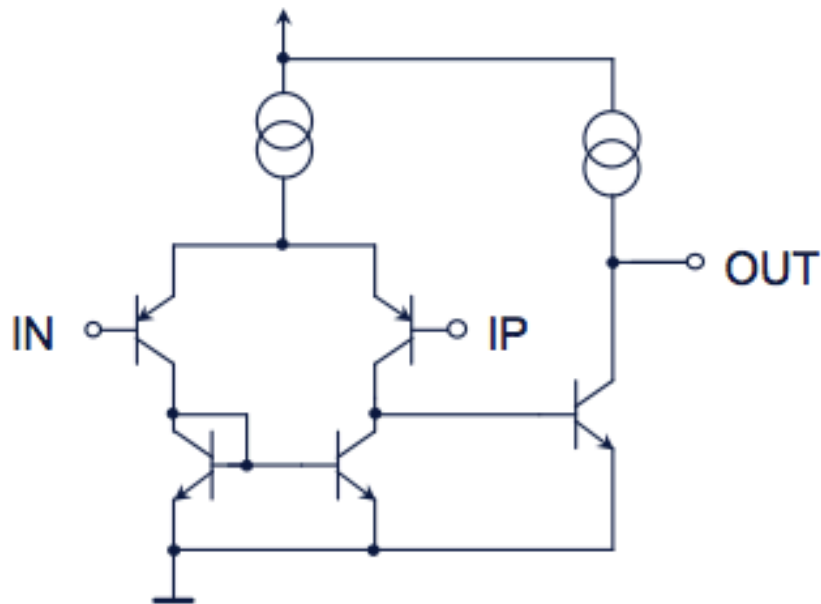
Realized in SPT (BCD) technology

Bipolar (BJT) - comparator

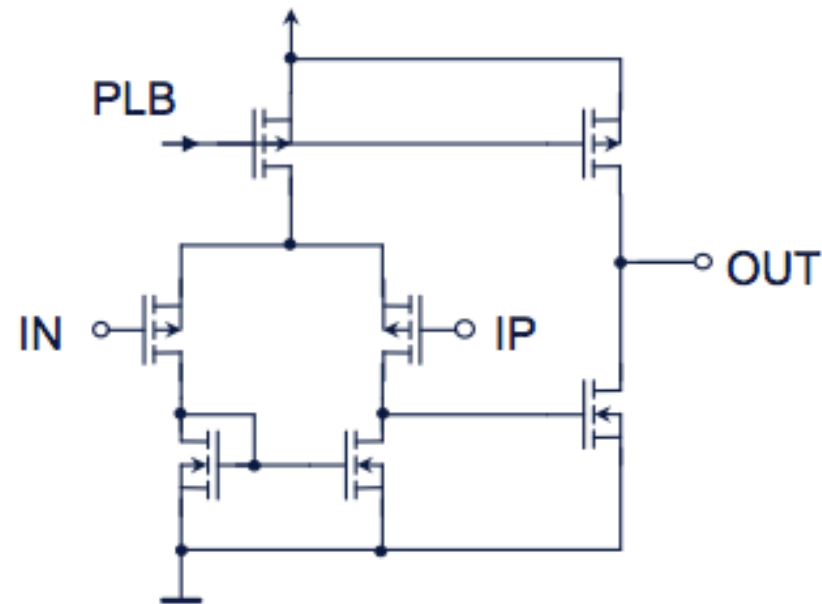


Using bipolar for the main components and MOS for the current sources is a good mix for bicmos opamps and comparators.

Comparator with P-device as input



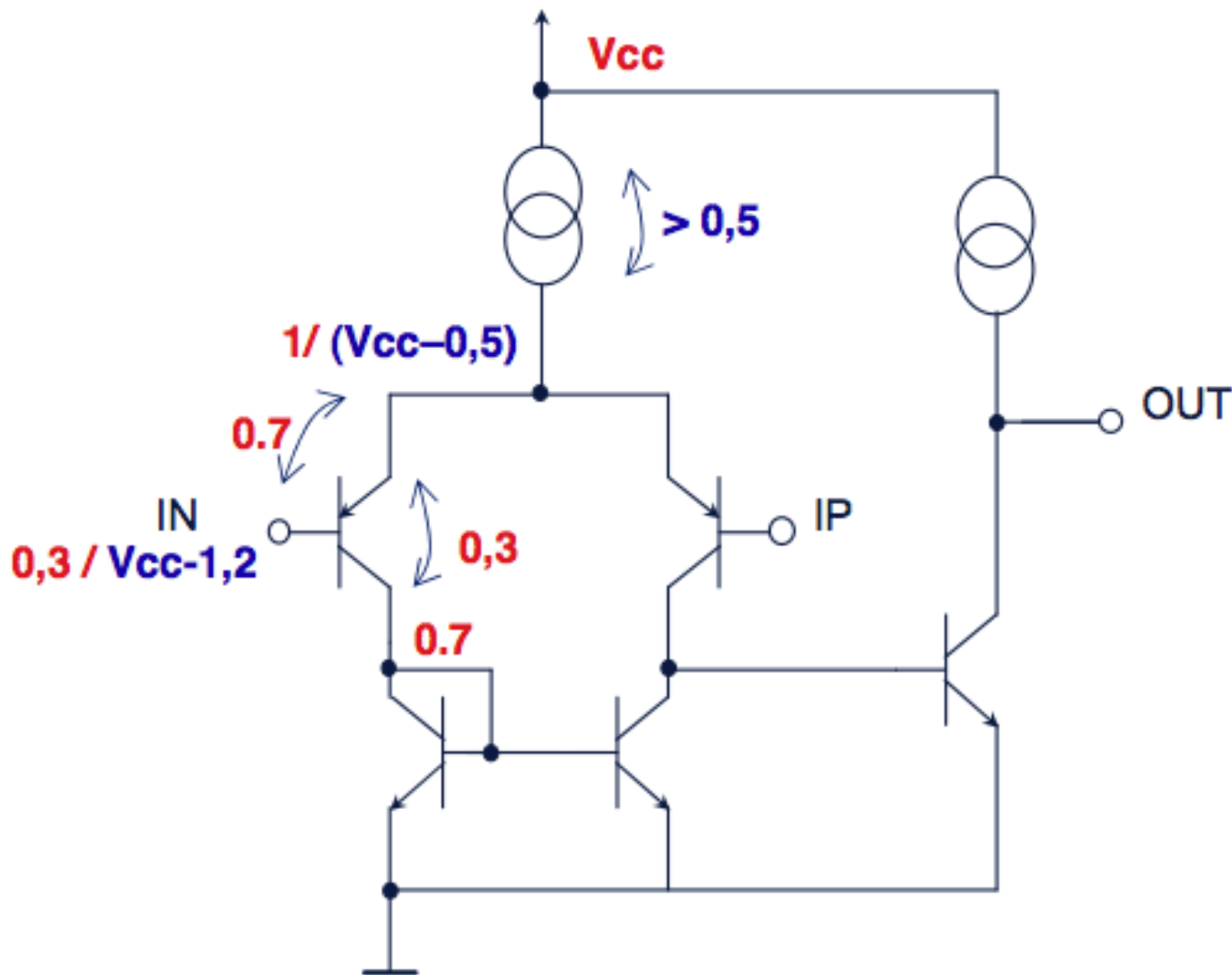
bipolar: npn-input



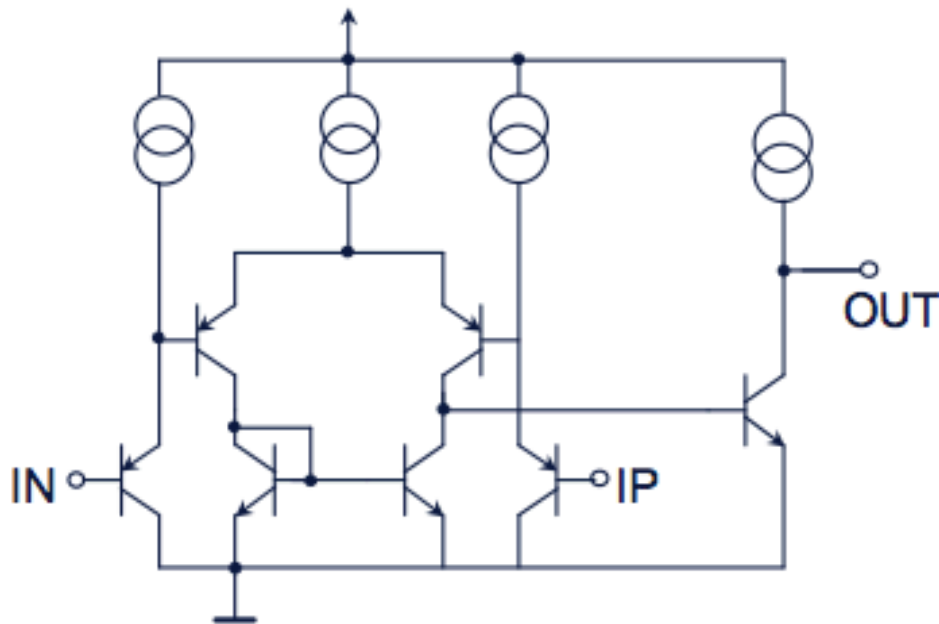
MOS: n-channel input

Use P-Input comparators for input voltages near GND (neg. supply voltage)

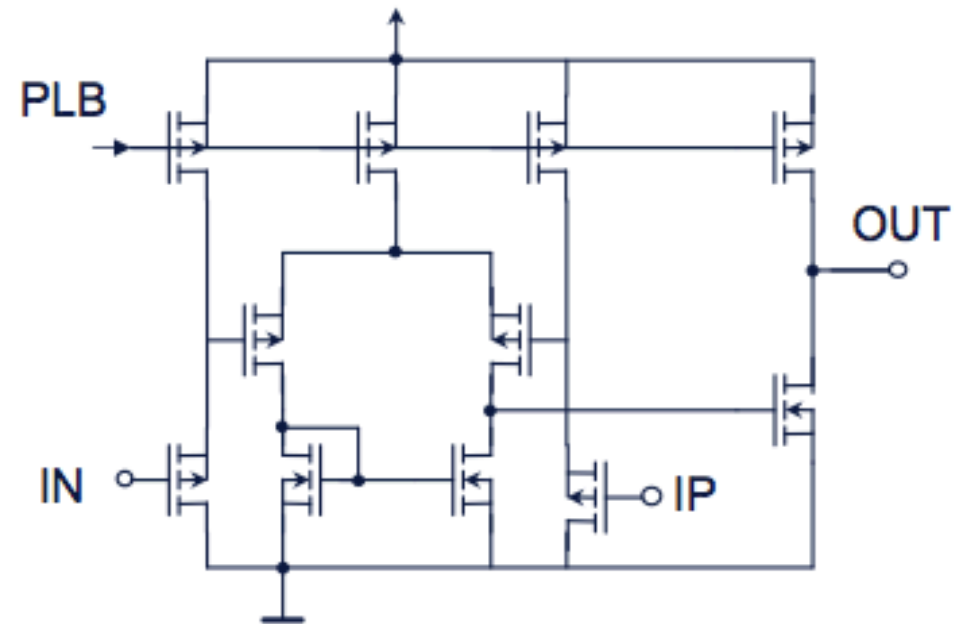
pnp-input Comparator: Input voltage range



Darlington inputs



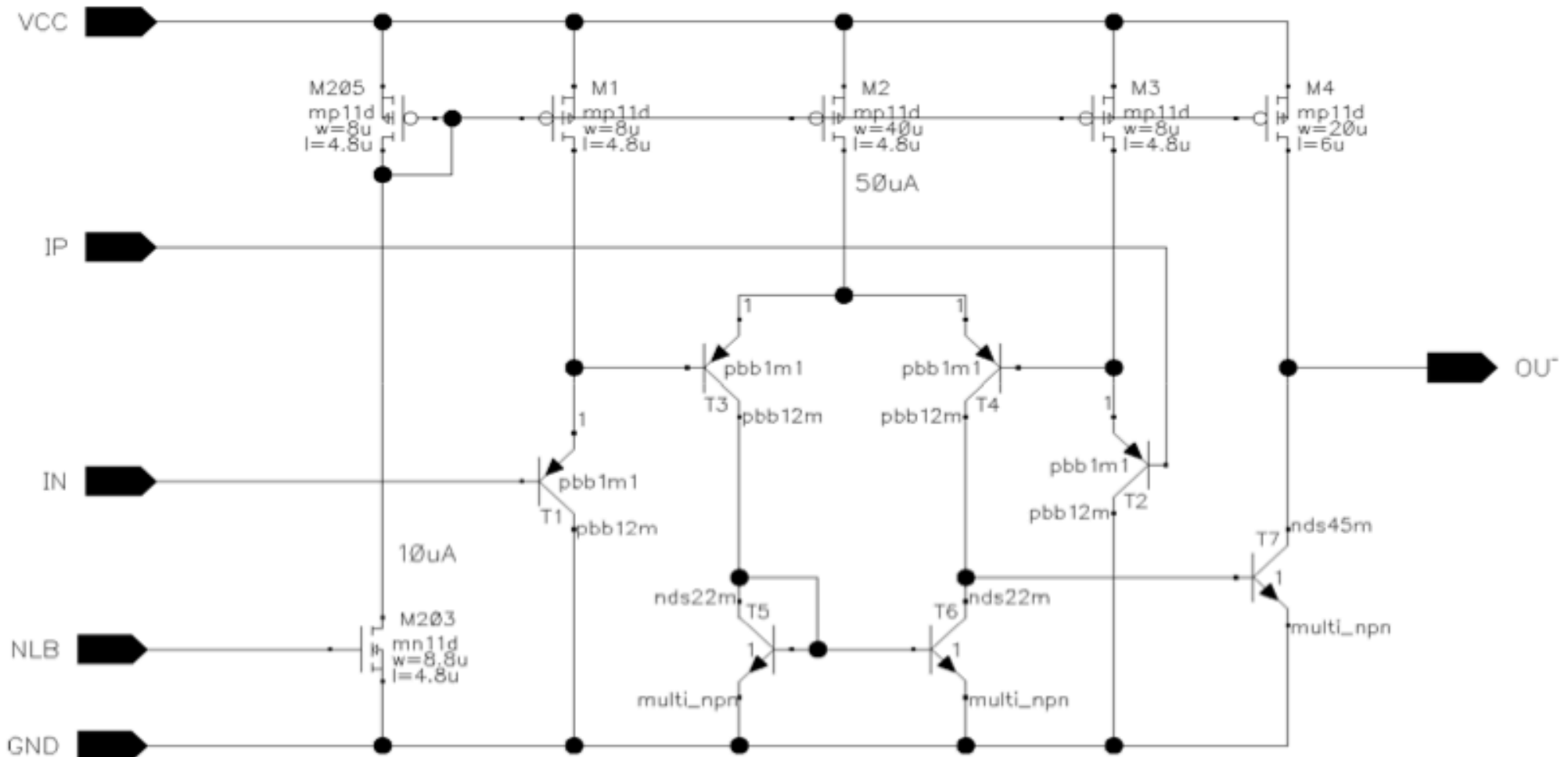
pnp - Darlington



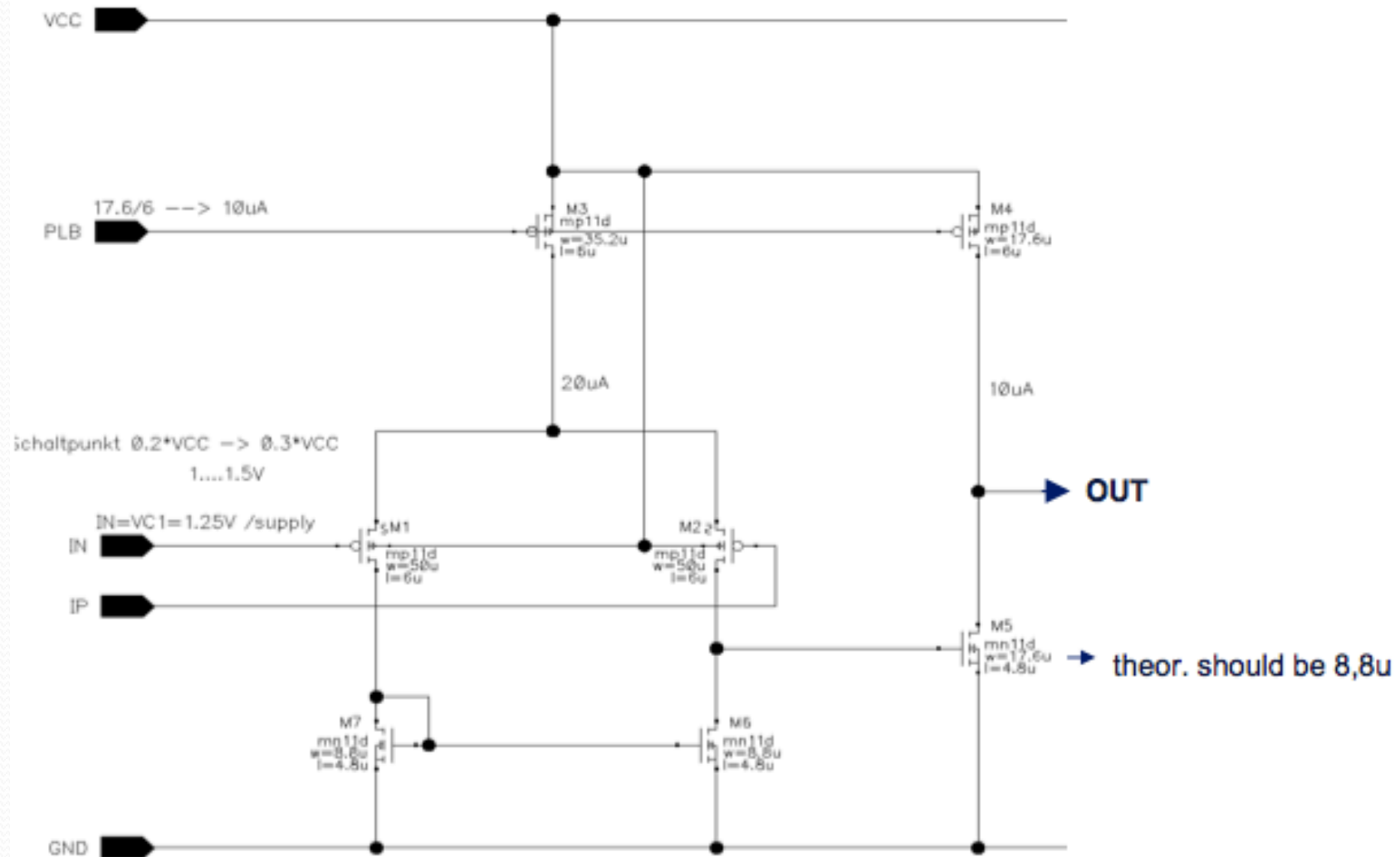
P-channel - levelshift

For input voltage range including GND use P-darlington input

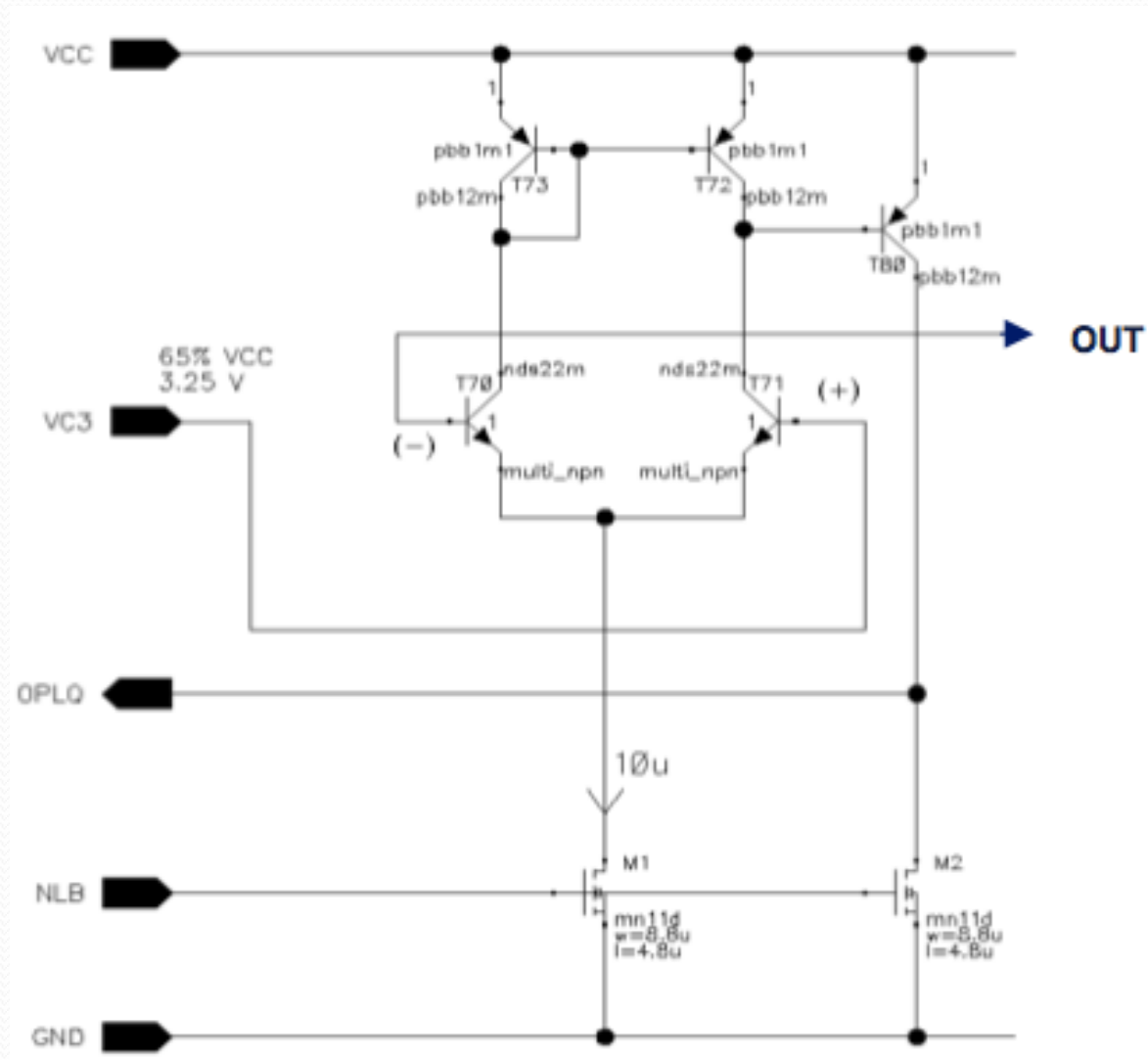
pnp darlington comparator: current measurement



P-input MOS comparator: logic input buffer with special input specification range



npn-input comparator: open load voltage measurement



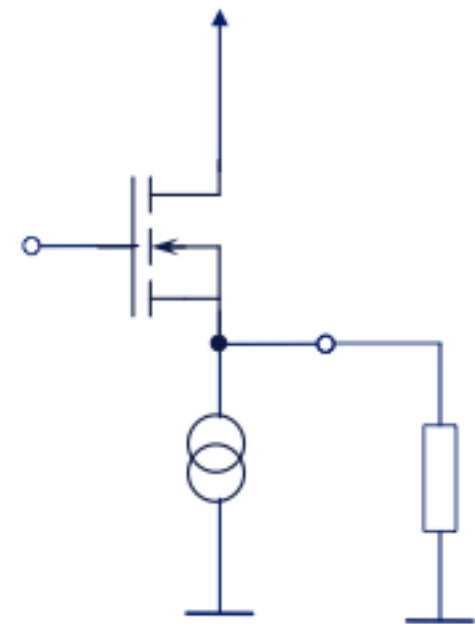
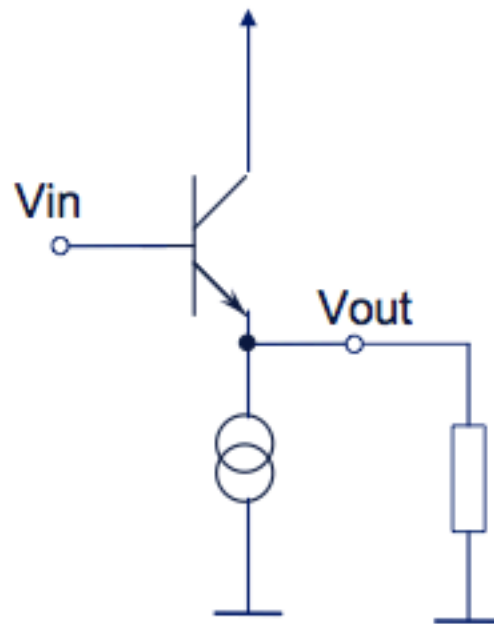
Emitter / Source Follower as Output Stage

Used as buffer.
No voltage gain ($=1$),
but can deliver
current into a load.

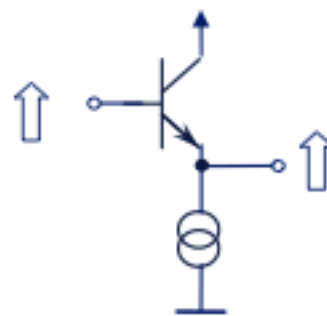
Does not invert the phase,
compared to gain stage

AC: $V_{out} = V_{in}$

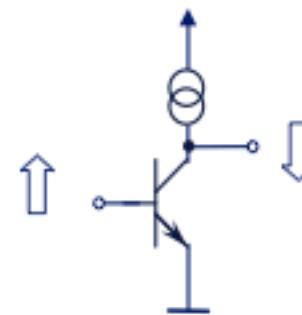
DC: $V_{out} = V_{in} - V_{be}$



signal polarity:

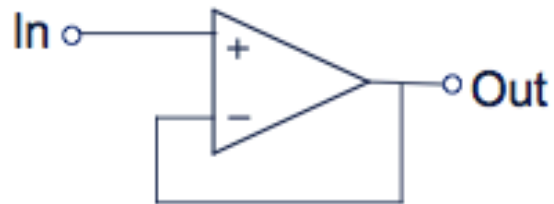


emitter follower
not inverting

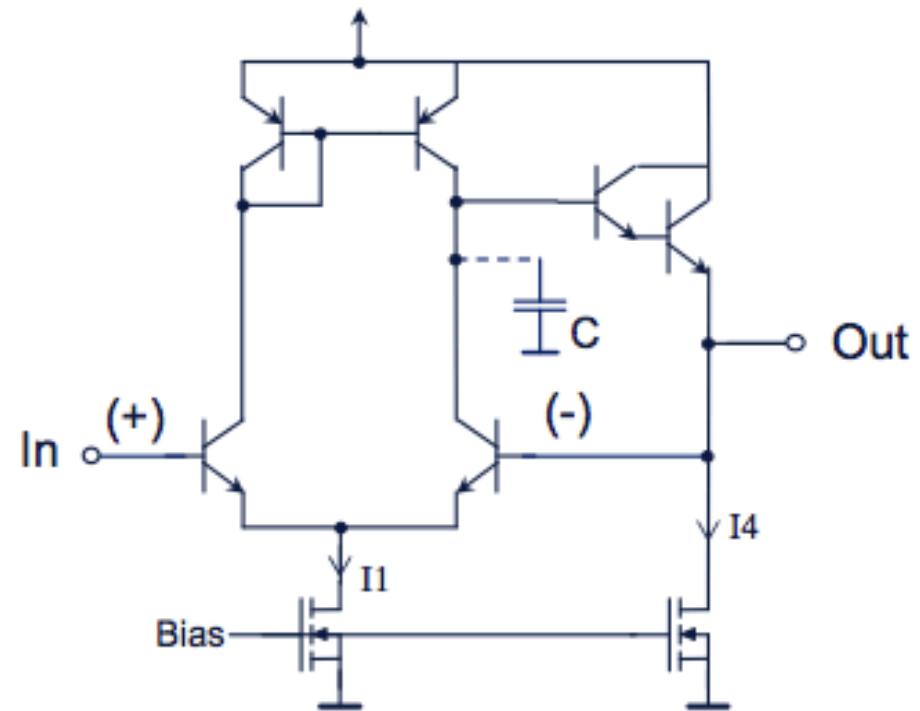


gain stage
inverting

Opamp as buffer

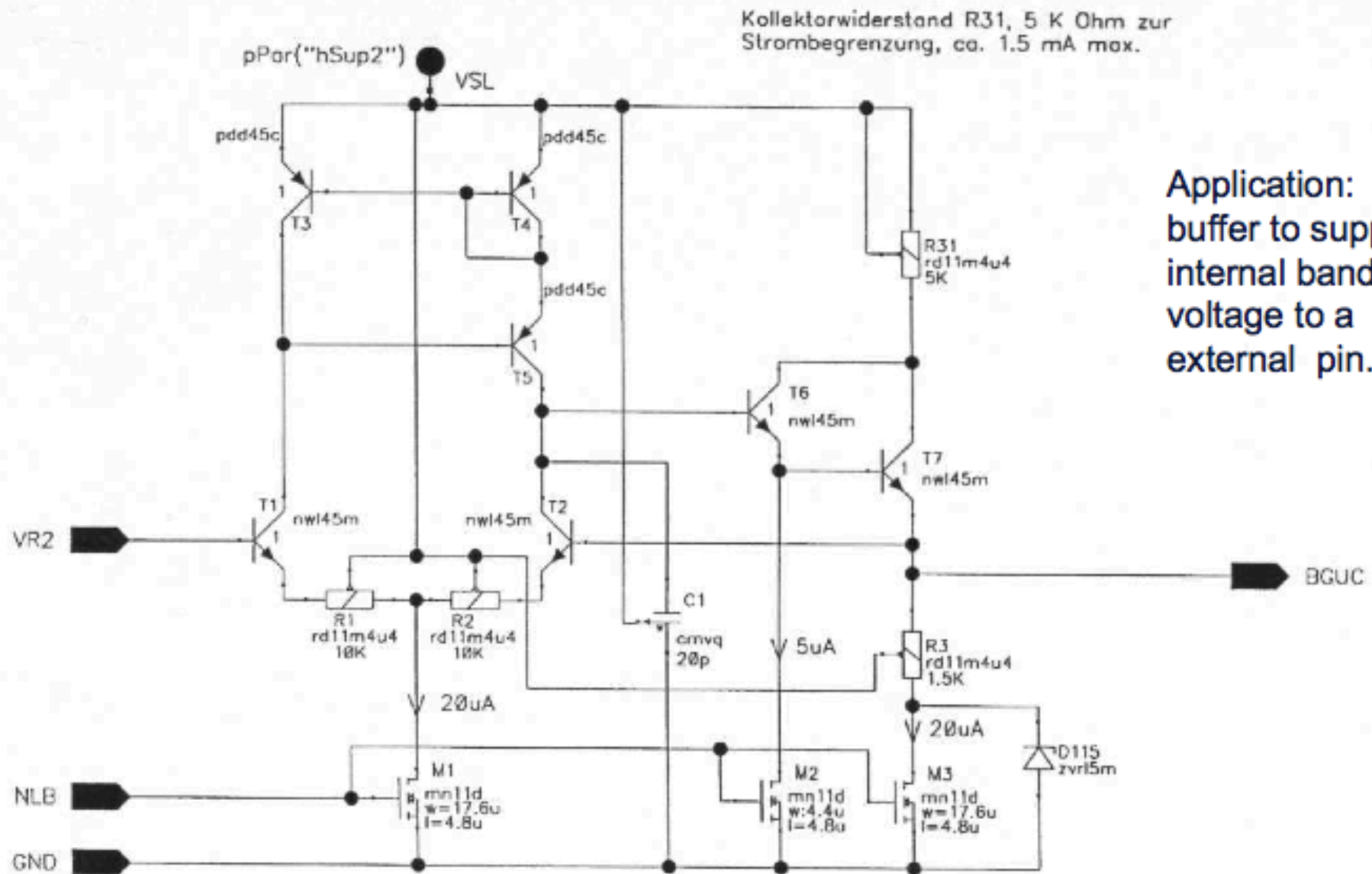


$V_{in} = V_{out} \rightarrow$ closed loop gain = 1

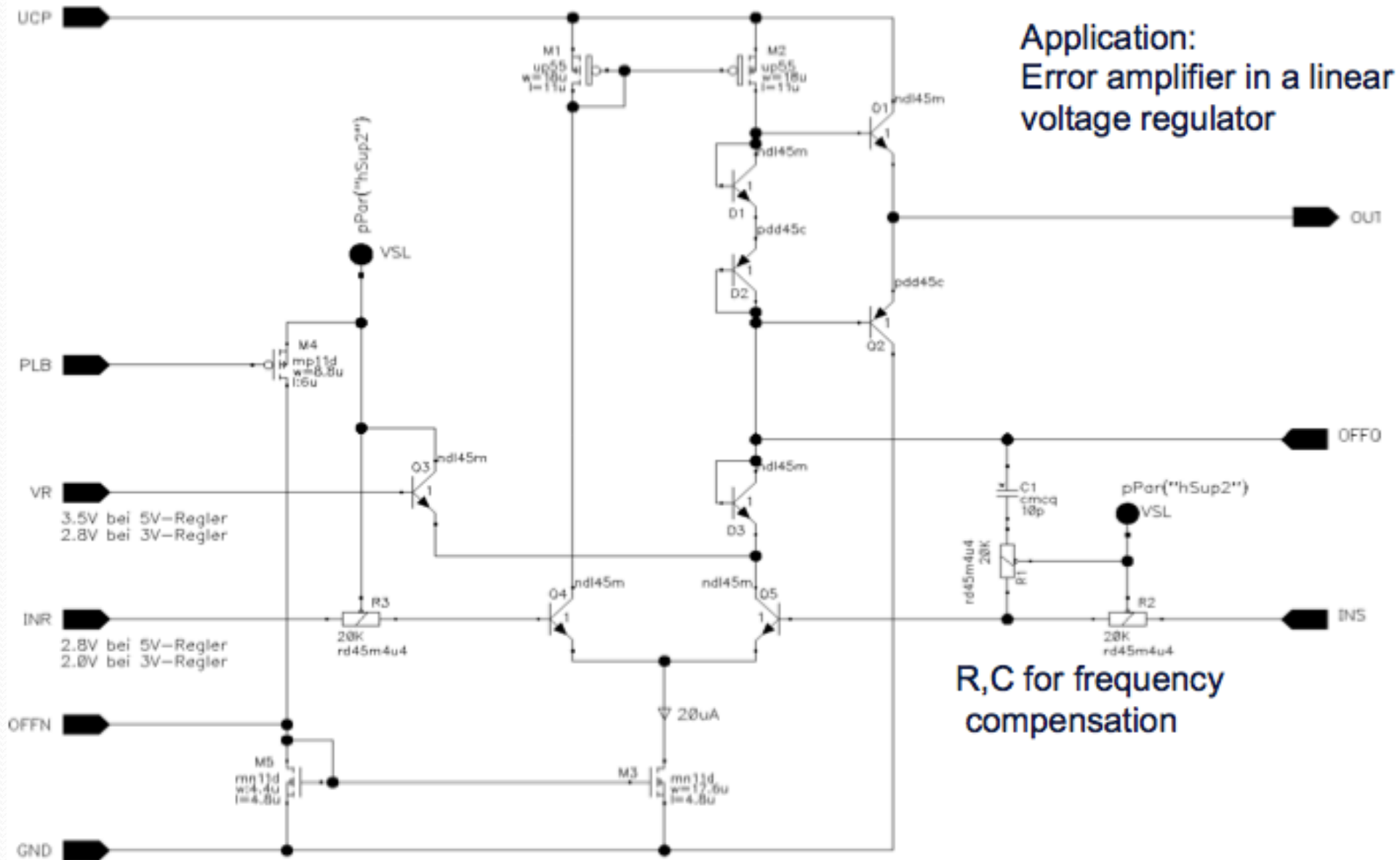


For a buffer-circuit with closed loop gain =1 an amplifier with one gain stage and an output buffer is good choice to simplify frequency response requirement for stability. Anyway check with AC simulation if compensation C is required.

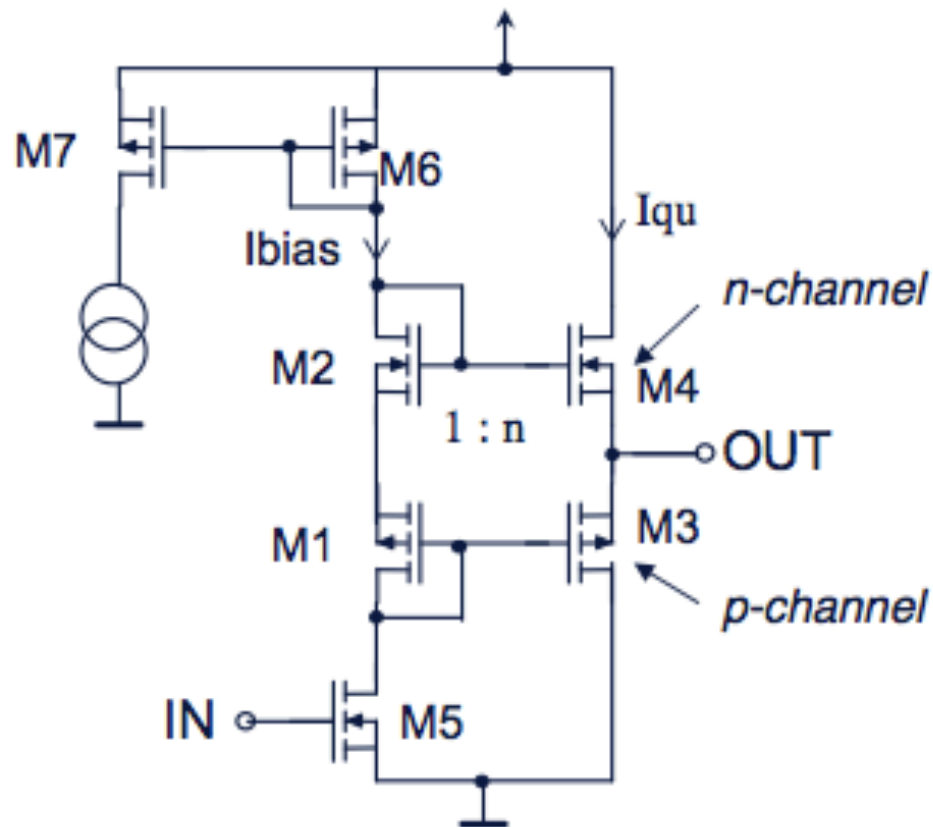
npn-input Opamp as buffer (voltage follower)



Opamp with push-pull output



MOSFET - Push-Pull Output Stage



Quiescent current (class A-operation) is defined by :

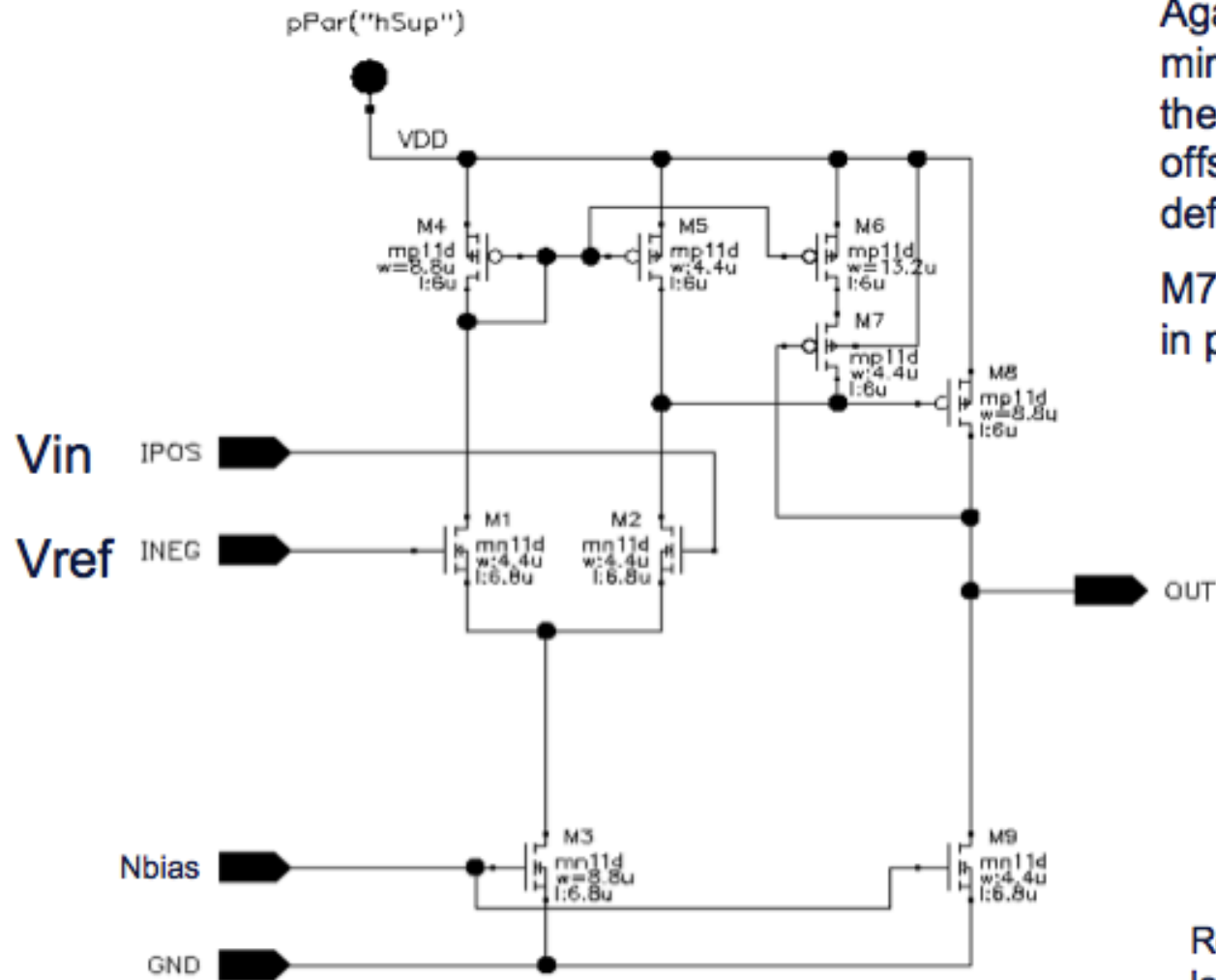
$$\frac{(w/l)_4}{(w/l)_2} = \frac{(w/l)_3}{(w/l)_1} = n$$

$$I_{qu} = n \cdot I_{bias}$$

Similar topology as bipolar push-pull.

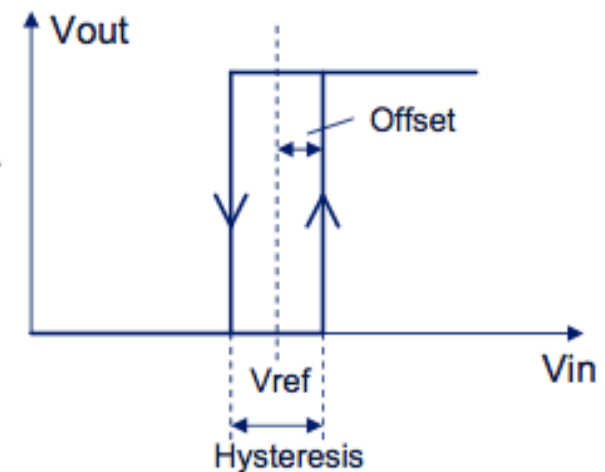
Current capability not so high caused by significant higher V_{gs} voltage of MOS compared to bipolar V_{be}

N-input MOSFET comparator with hysteresis



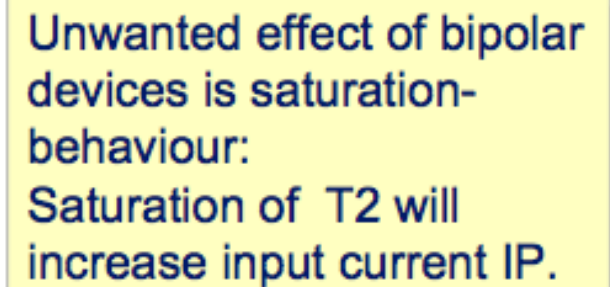
Against the rules here the current mirror transistors $M4/M5$ has not the same size! This causes an offset and this offset is used to define a hysteresis.

$M7$ is used as a switch to connect in parallel $M6$ to $M5$ if V_{out} is low.



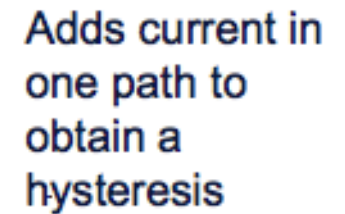
Resulting offset is larger if gain of comp. is low, therefore w of $M1$, $M2$ are small

Bipolar input for low offset
MOS for smaller area

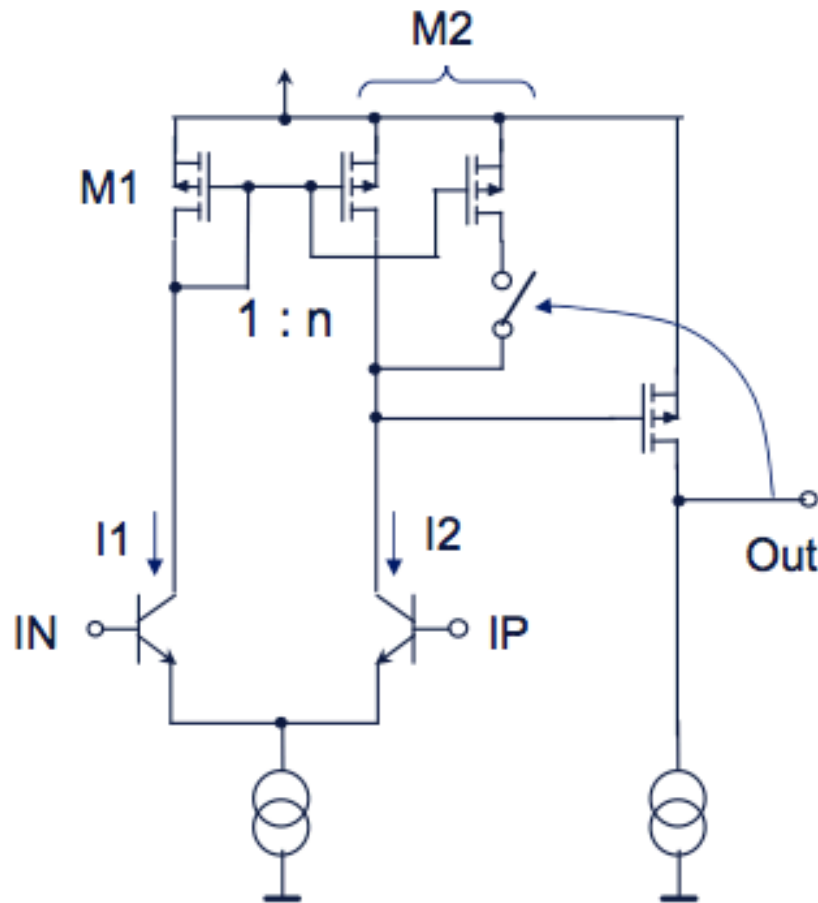


IN BICMOS designs be careful at the transition points bip-MOS to avoid bipolar saturation effects

Q3 clamps V_c of Q2
not to go lower than V_b



Calculation of hysteresis of bipolar comparator

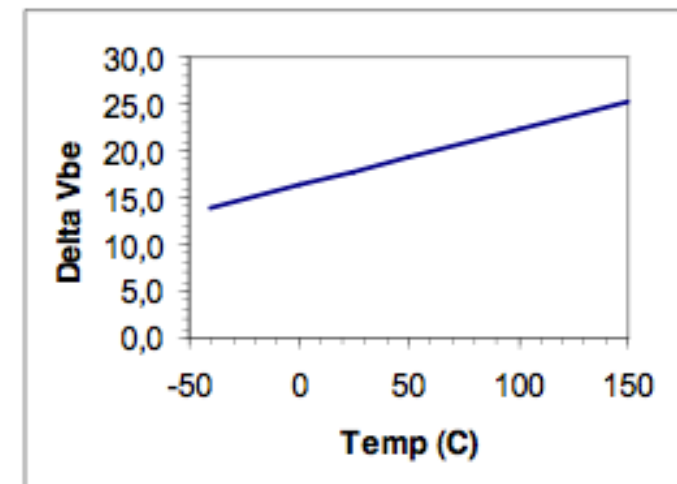


If size of M1 not equals size of M2, currents I_1 , I_2 will not be equal. This causes an offset which can be calculated with the V_{BE} equation for bipolar:

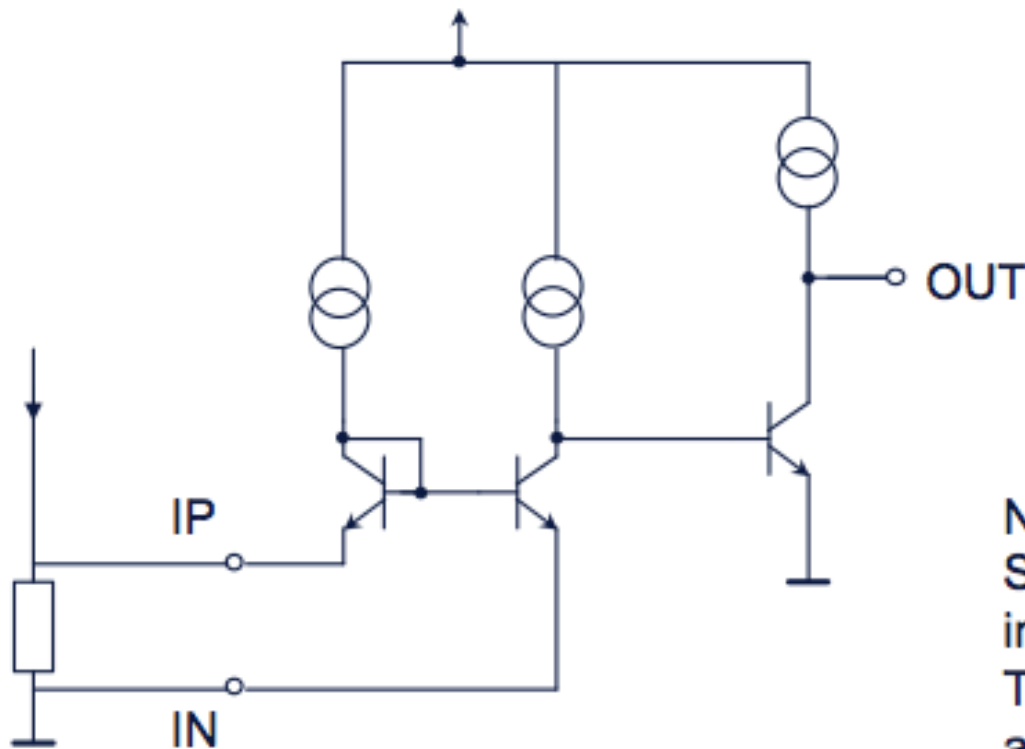
$$\Delta V_{BE} = \ln\left(\frac{I_1}{I_2}\right) \cdot V_t$$

Switching on and off this offset depending of the output voltage results in a hysteresis. Hysteresis will be temperature dependent according V_t behaviour

example
for $n=2$:



Comparator with emitter-input

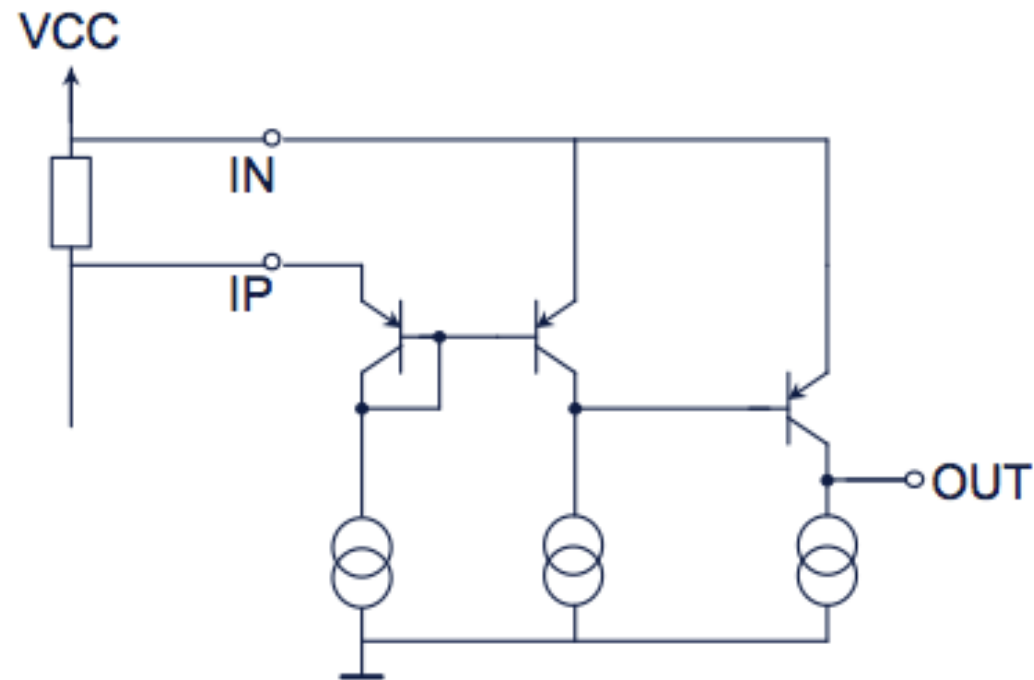


Note:
Supply current is equal to
input currents.
To take into account for
accuracy (offset)

npn- emitter-input

For input-range near ground e.g. current measurement

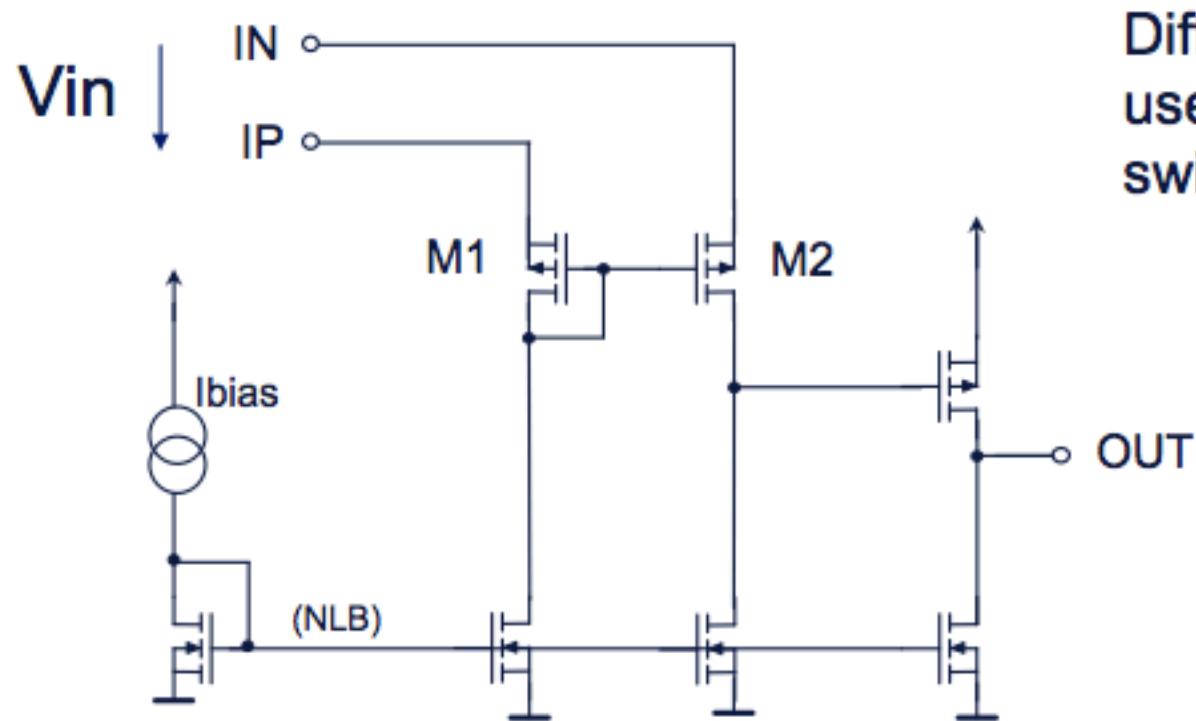
pnp with emitter - input



for input-range near V_{CC}

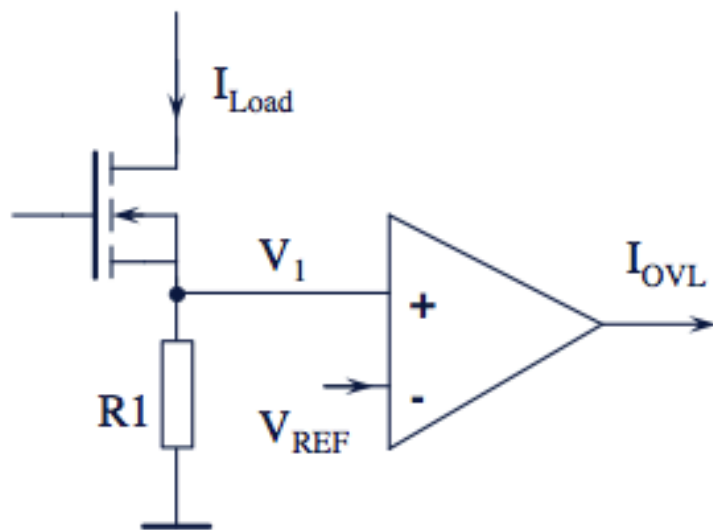
e.g. current measurement for high side switches

P-channel source-input comparator

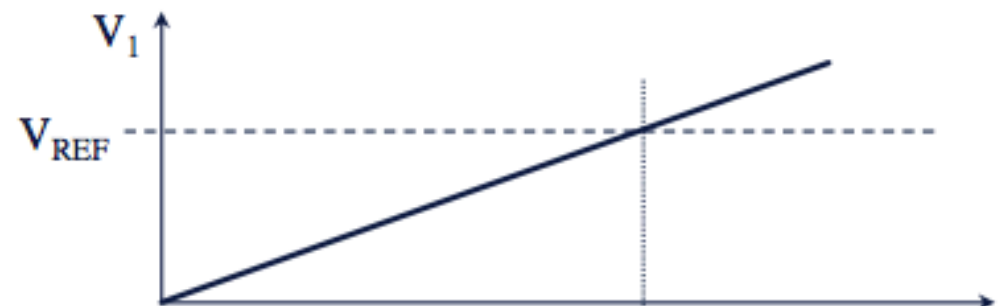


Different size of M1, M2 can be used to design a wanted switching threshold $V_{in} > 0$

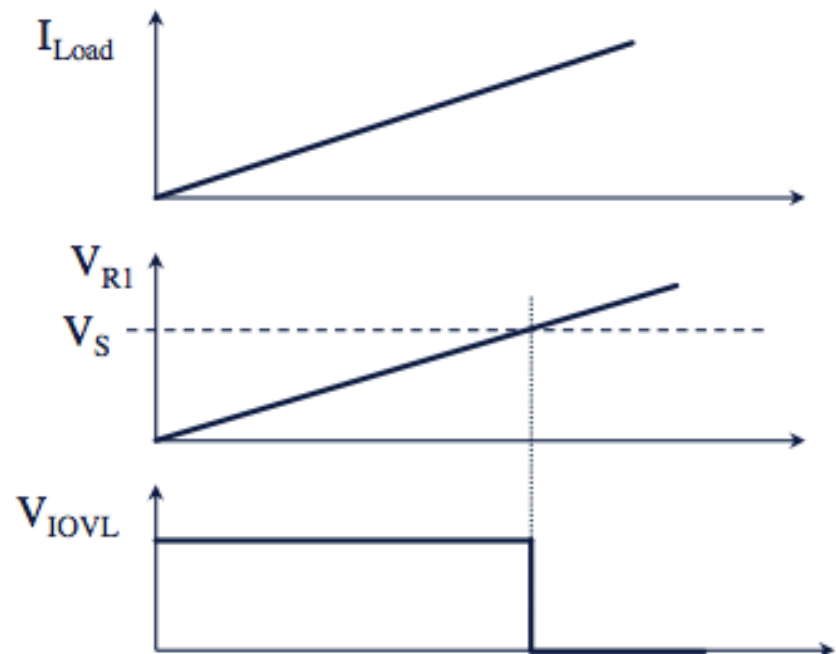
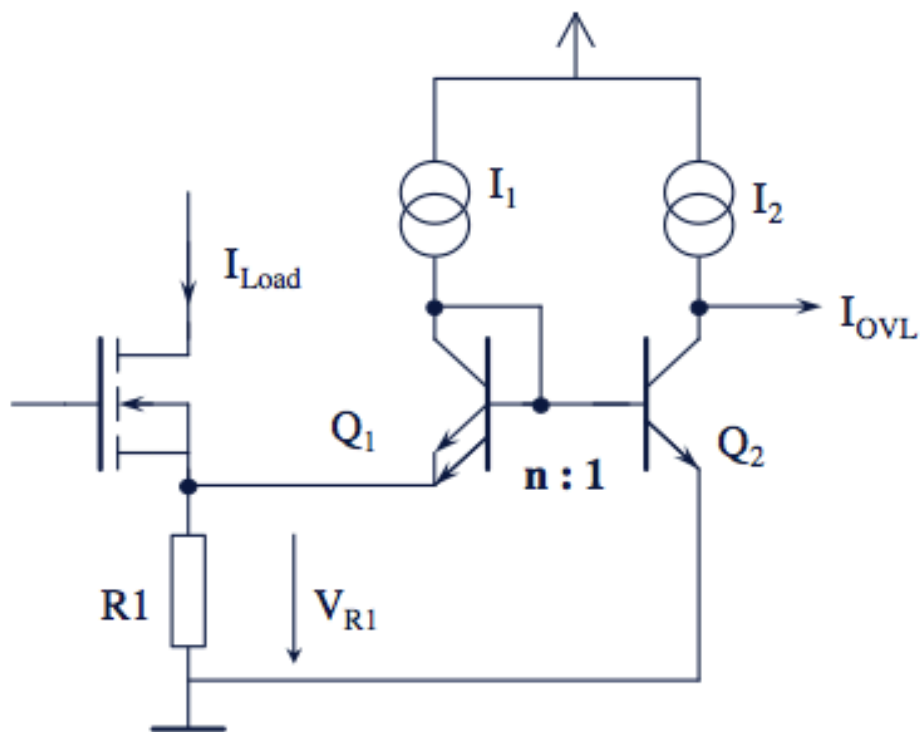
Current measurement with Shunt-Resistor



Requires a reference voltage



Current Measurement with Shunt-Resistor: emitter-input-comparator, uses ΔV_{BE} principle



Calculation of threshold V_S ::
$$V_S = V_T \cdot \ln\left(n \cdot \frac{I_2}{I_1}\right)$$

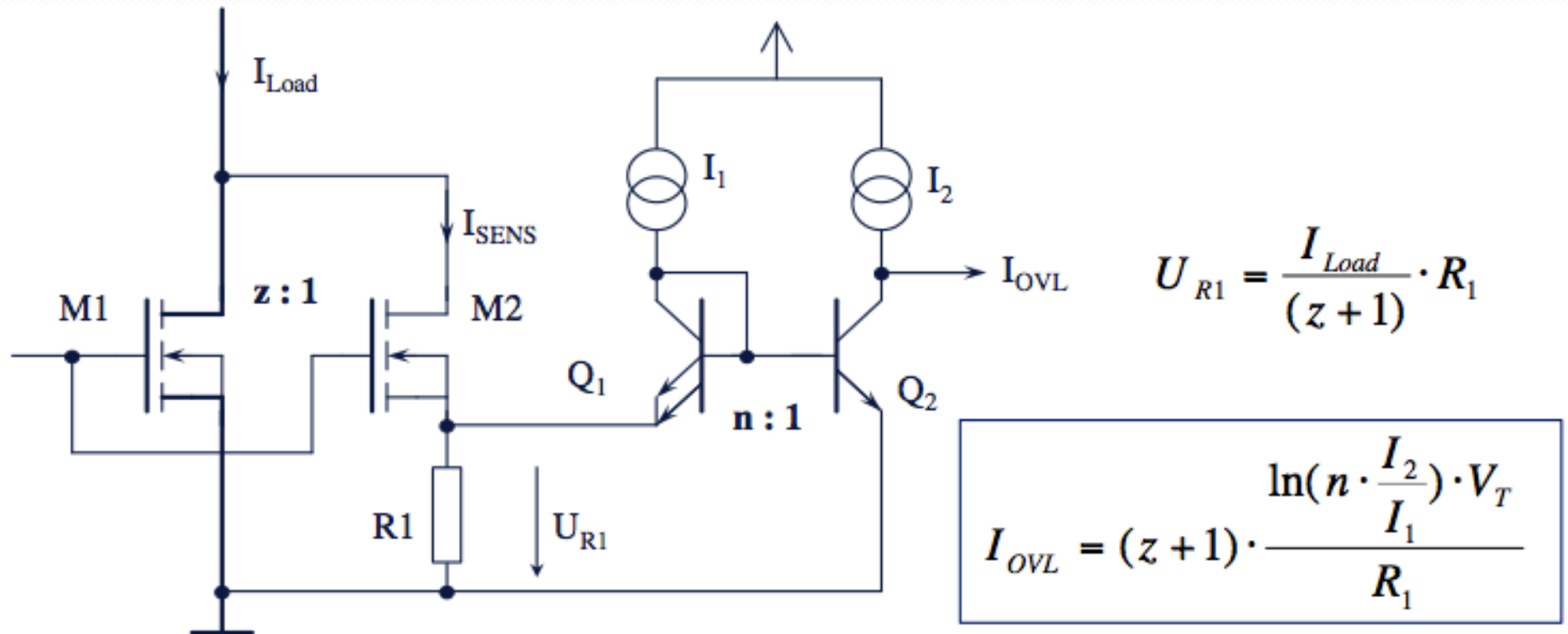
n = ratio of emitter areas Q_1/Q_2

e.g. $I_1=I_2$, $n=2 \Rightarrow V_S = 18 \text{ mV @ } 25\text{C}$

$R1$ as aluminum-resistor results in a good first order temperature compensation of V_T ($TC_{AL} = 3,8 \cdot 10^{-3}$)

Current Measurement with Sense-Transistor

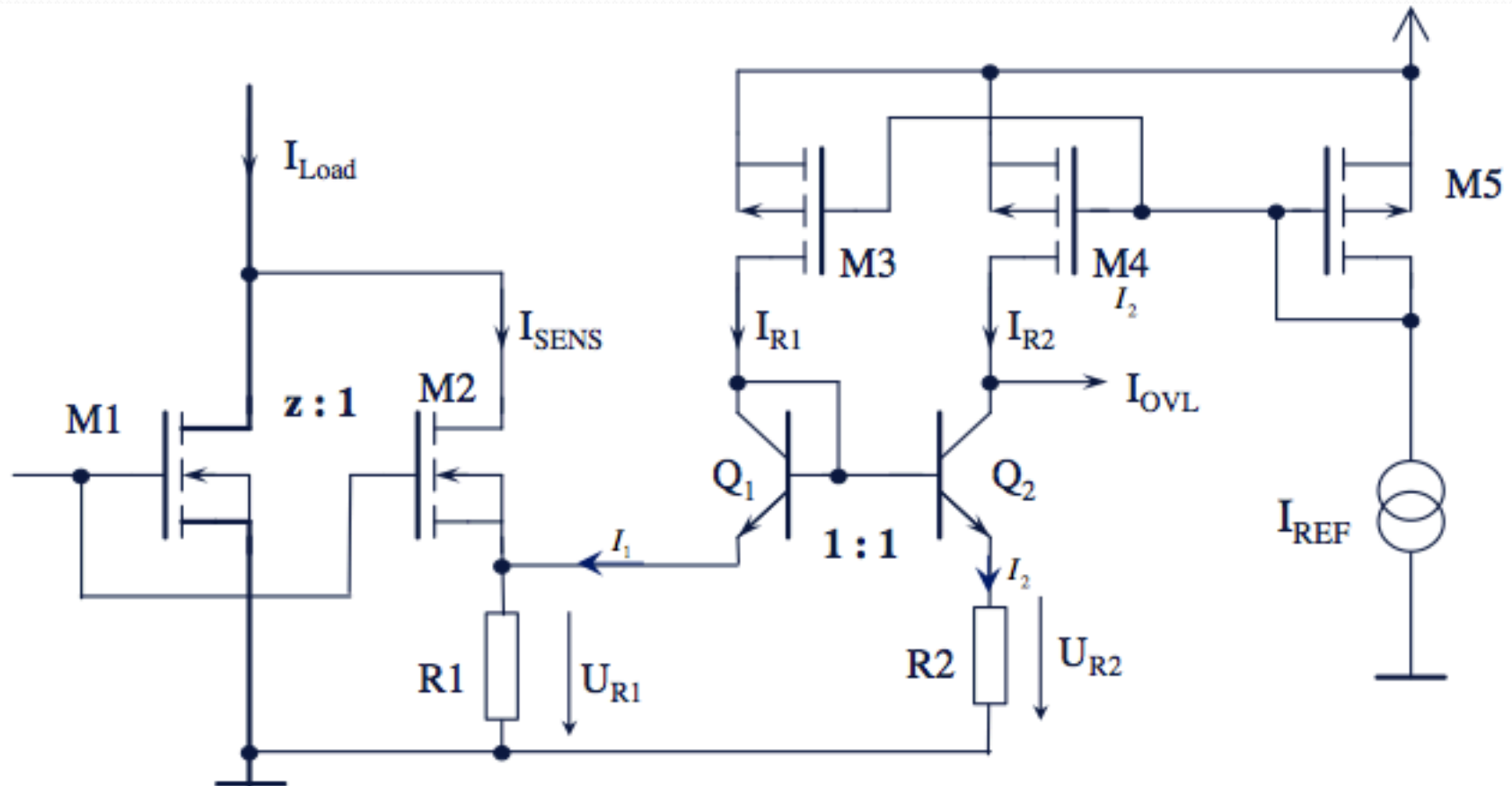
Shunt-resistance in series to sense-cell



The layout of the powerdevice (DMOS) in general consist of many cells (> 1000). One or few cells are used as sense cells, the current ratio between main ($M1$) and sense DMOS ($M2$) is defined by the ratio z . The shunt resistor $R1$ must therefore not be dimensioned for the full load current and, as additional advantage, the shunt resistor does not generate a voltage drop in series to the power DMOS.

Current Measurement with Sense-Transistor

Sense-cell with Shunt-Res., Reference-Current with Reference-Resistor



$$U_{R1} = \left(\frac{I_{Load}}{z+1} + I_1 \right) \cdot R_1$$

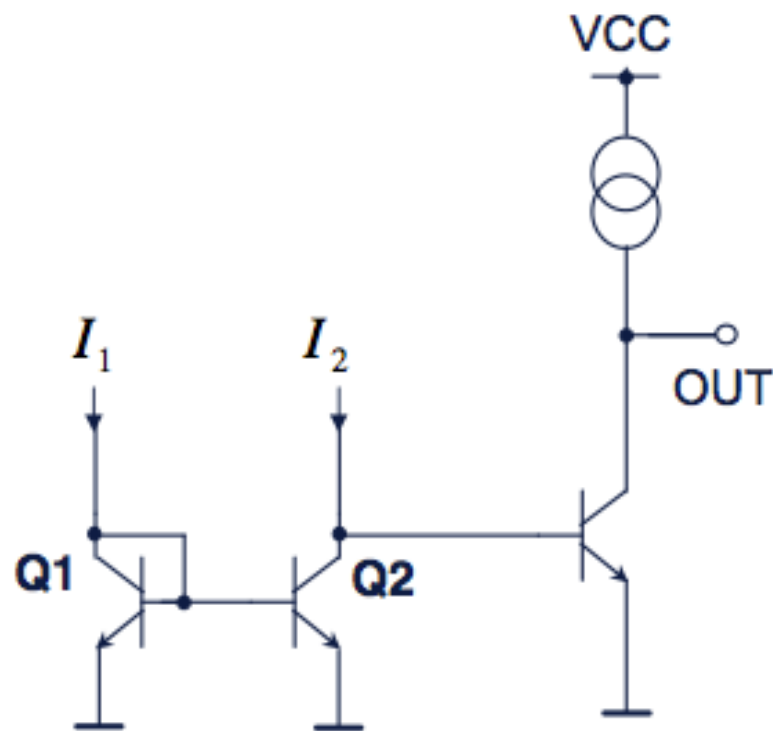
$$U_{R2} = I_2 \cdot R_2$$

with $I_1 \ll I_{sens}$ and $I_1 = I_2 = I_{REF}$
the threshold can be calculated:

$$I_{OVL} = I_{REF} \cdot (z+1) \cdot \frac{R_2}{R_1}$$

$Q1, Q2$ are input stages of a comparator and do not use the delta VBE principle!

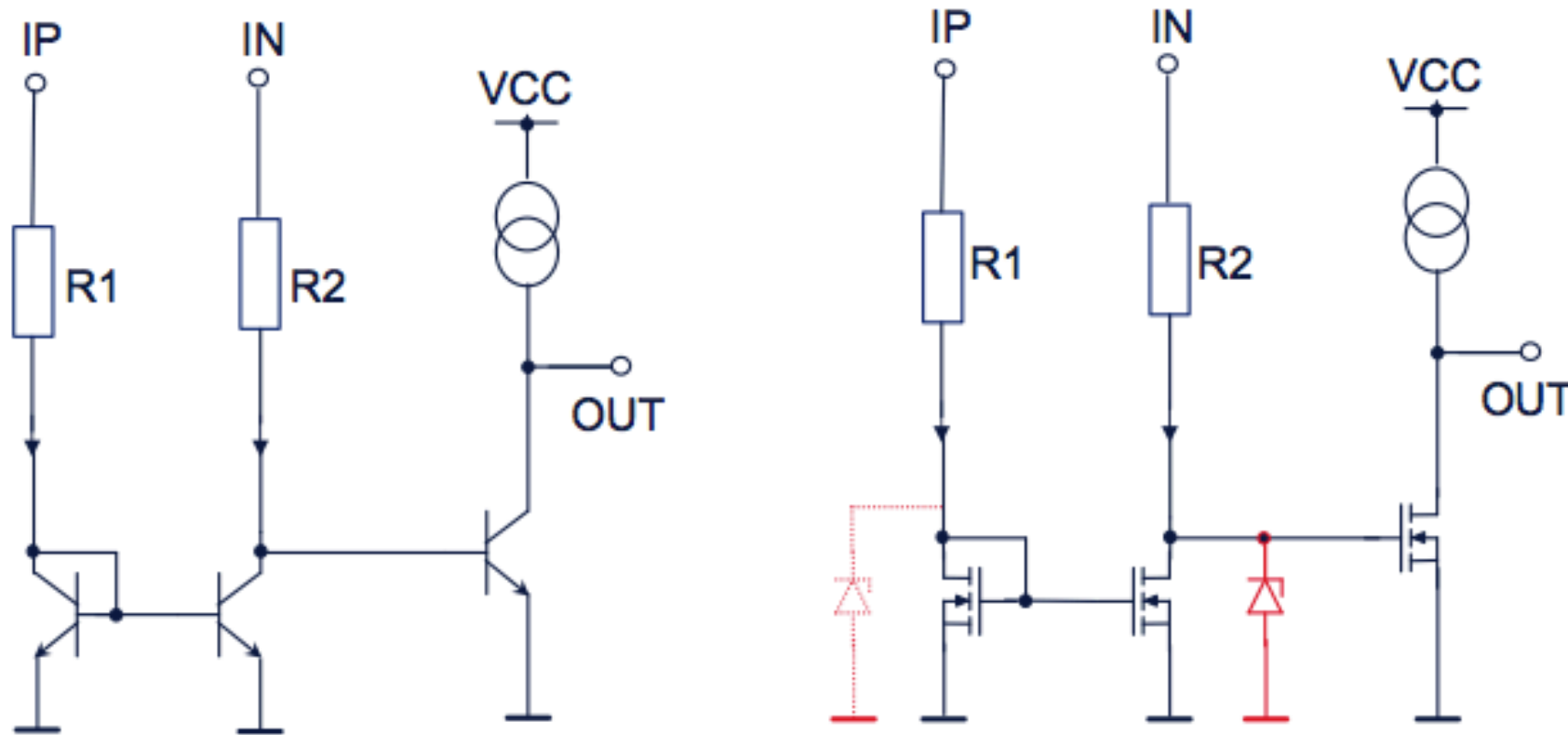
Current Comparator



A current mirror (bip or MOS) can simply be used to compare two currents

You can call it a „current amplifier“ or a „current comparator“ where the input signals are not voltages, but currents.

High Voltage Input - Current Comparator with low voltage devices



Only resistors have to withstand the high input voltage (no problem with poly resistors), active devices can be low voltage devices.

MOS-circuit: take care to add z-diodes for overvoltage clamping e.g. against ESD events



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